

Introduction to Gravitational Waves and Pulsar Timing Array



Reading Project Report

Submitted by

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Chapter 1

Introduction to Pulsars

What are pulsars?

Pulsars are magnetised neutron stars that appear to emit periodic short pulses of radio radiation (electromagnetic waves at the lowest-frequency, longest-wavelength end of the electromagnetic spectrum).

The time period of their oscillations is between 1.4 ms and 8.5 s. A neutron star is one with a very small radius and an extremely high mass. As a result, the density of one is very high.

In the event of such high pressure as a supernova, the protons and electrons of the atoms involved undergo a phenomenon called inverse beta decay.

Inverse beta decay:

$$p + e^- \longrightarrow n + \nu_e \quad (1.1)$$

Pulsars are not literally ‘pulsating’. The neutron stars that rotate (at very high speeds) emit beams of radiation, which are detected by the observer, and cause a **lighthouse effect**. Even though the mechanism of this radiation is not well known, it has various applications.

Applications of neutron stars and pulsars include **using it to study extreme conditions of matter** (The extremely high gravitational potential, density (more than that of atomic nucleus), and magnetic field strength), and **using it as a clock** (Pulse periods are very accurate. So much so that it permits sensitive measurement of precision relativistic observables).

1.1 Neutron star masses and densities

What really causes these pulses though? It is the combination of various factors.

Rapid rotation of the neutron star with a strong magnetic field. A misalignment between the magnetic axis and rotation axis.

For a pulsating star, density controls the pulsation period. The gravity pulls the star inwards, while the pressure pushes it out. By this, the speed of oscillation depends on the the gravity per unit area.

Higher density $\rightarrow$ strong gravity $\rightarrow$ faster oscillations

Lower density $\rightarrow$ weaker gravity $\rightarrow$ slower oscillations

The period of oscillation for sun is in hours or days. But the observed period of the pulsars is in the range of ms to s. So the speed of oscillations has to be much faster, which means the star has to be of higher density. The neutron star fits perfectly in the calculations (it is over two hundred trillion times denser than the sun).

Mathematically, the condition that we can apply here, considering how density controls the period of oscillation: The centrifugal acceleration at its equator does not exceed the gravitational acceleration.

$$F_{centripetal} < F_{gravitational}$$

$$(acc^n)_{centripetal} < (acc^n)_{gravitational}$$

$$\omega^2 R < \frac{GM}{R^2}$$

Substitute and write the angular acceleration ' ω ' in terms of 'T'

$$(\frac{2\pi}{T})^2 R < \frac{GM}{R^2}$$

Simplifying the equations further,

$$T^2 > (\frac{4\pi^2}{GM}) R^3$$

$$T^2 > (\frac{4\pi R^3}{3}) \frac{3\pi}{GM}$$

Writing this in terms of mean density (ρ) ie. $\frac{M}{\frac{4}{3}\pi R^3}$

$$\frac{M}{(\frac{4}{3}\pi R^3)} > \frac{3\pi}{GT^2}$$

$$\rho > \frac{3\pi}{GT^2}$$

$$T = (\frac{3\pi}{G\rho})^{1/2} \quad (1.2)$$

Chandrashekhar Mass:

A dense star is kept intact due to the presence of the balancing forces, **the gravity** (tries to collapse the star) and **the electron degeneracy pressure** (the pressure coming due to the intense packing of the electrons)

As the mass of star increases, the electron degeneracy pressure no longer stays enough to be equal to the gravitational attraction. This limit is called the **Chandrashekhar limit**.

$$M_{Ch} = 1.4M_o \quad (1.3)$$

Substitute the values for the boundary condition of mass equal to 1.4 times that of sun. The first ever neutron star to be discovered had a time period of 1.4×10^{-3} seconds. For such a star, the radius comes out to be equal to around 20 km. Thats it!

This is how dense these objects really are..!

Canonical Neutron stars:

These are objects with a mass of around 1.4 times that of the sun, and radius of approximately 10 km. The moment of inertia of such an object (given solid sphere) will be:

$$I = \frac{2}{5}MR^2$$

for R = Radius of the sun and $M = 1.4$ times mass of the sun.

$$I = \frac{2}{5}(1.988 \times 10^{30})(1.4 \times 696342)^2$$

$$I = 0.76 \times 10^{41} kgm^2 \quad (1.4)$$

The Chandrashekhar limit is the lower limit for the existence of neutron stars. The upper limit can be said equivalent to 3 times the mass of sun. If the upper limit is crossed, the neutron star must collapse into a black hole.

Equation of States (EoS):

The equation of states tell us how the pressure is dependent on the density. Gravity tries to collapse the star while internal pressure resists collapse. The EoS determines how much mass can be supported.

$$1.4M_o < M < 3M_o \quad (1.5)$$

1.2 Magnetic field and Dipole Radiation

Magnetic Fields:

The stars are said to have an approximate magnetic dipole (every massive object does, take the example of our Earth. Magnetic dipoles exist, and so there are magnetic field lines present around the planet).

Now we are talking of compressing this massive star into a small object (say $10^6 km$ to $10km$ radius), the cross sectional area reduces by 10^{10} times ($\propto r^2$). So a young neutron star shall have massive dipolar fields. Conservation of angular momentum also confirms that, the smaller the radius, the faster the spin.

Magnetic Dipole Radiation:

Say, the magnetic dipole moment is perfectly aligned with the rotation axis of the neutron star. Then the magnetic field pattern remains the same, and there is no emission of any electromagnetic radiation.

This is an imaginary case. Let that get tilted by an angle α . Larmor's equation: Any charged particle radiates when accelerated and that the total radiated power is proportional to the square of the acceleration. Mathematically,

$$P = \frac{2}{3} \frac{q^2 \dot{v}^2}{c^3} \quad (1.6)$$

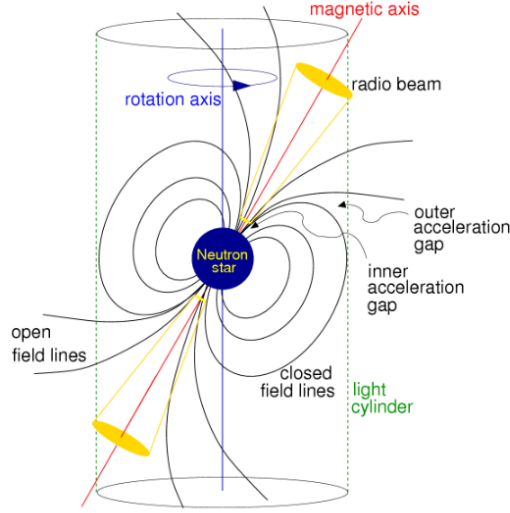


Figure 1.1: Magnetic field lines around a neutron star

Rewriting the Larmor's formula in terms of rotating electric dipole,

$$P = \frac{2}{3} \frac{(q\ddot{r} \sin\alpha)^2}{c^3}$$

$$P = \frac{2}{3} \frac{\ddot{p}_\perp^2}{c^3} \quad (1.7)$$

where $m_\perp = m \sin\alpha$ is the perpendicular component of the magnetic dipole moment.

1.3 Spin-Down Luminosity

The term 'spin down' refers to the neutron star **losing** energy as it spins. This indicates the amount of power the pulsar is releasing into its surroundings. It is the **rate of slowing down of the star**.

The rotational kinetic energy E of a spinning object is given by

$$E = \frac{1}{2} I \omega^2$$

$$E = \frac{I}{2} \left(\frac{2\pi}{T} \right)^2$$

$$E = \frac{2I\pi^2}{T^2} \quad (1.8)$$

The moment of inertia of a solid spherical object of mass M and radius R is given by

$$I = \frac{2}{5} MR^2$$

Now from equation (1.4),

$$I = 0.76 \times 10^{41} \text{kgm}^2$$

We plug in the values for I and $T = 0.033\text{s}$ of a crab pulsar.

$$E = 1.8 \times 10^{49} \text{ergs} \quad (1.9)$$

Now we try to determine the rate at which the neutron star is losing energy. Moving to equation (1.8),

$$E = \frac{1}{2} I \omega^2$$

$$-\dot{E} = -\frac{d}{dt} \left(\frac{1}{2} I \omega^2 \right) = -I \omega \dot{\omega}$$

This quantity $-\dot{E}$ is called the **spin down luminosity**.

Substitute $\omega = \frac{2\pi}{T}$

$$-\dot{E} = -I \left(\frac{2\pi}{T} \right) \left(\frac{\dot{2\pi}}{T} \right)$$

$$-\dot{E} = \frac{4\pi^2 I}{T^3} \dot{T} \quad (1.10)$$

The Crab pulsar has $T = 0.033\text{s}$ and $\dot{T} = 10^{12.4}$. If $I = 10^{45} \text{gcm}^2$, the neutron star is losing rotational energy at that rate:

$$-\dot{E} = 4 \times 10^{38} \text{ergs}^{-1} \quad (1.11)$$

Minimum Magnetic Field Strength:

We have so far seen, that the neutron star loses energy through magnetic dipole radiation. There should be a minimum magnetic field originally, below which the star cannot afford to lose its field anymore. This minimum is the minimum magnetic field strength of that star.

Mathematically,

$$P_{rad} = -\dot{E}$$

Substituting the values for each of the sides,

$$\frac{2}{3c^3} (BR^3 \sin\alpha)^2 \left(\frac{4\pi^2}{P^2} \right)^2 = \frac{4\pi^2 I \dot{P}}{P^3}$$

$$B = \left(\frac{3c^3 I}{8\pi^2 r^6} \right)^{1/2} \frac{\sqrt{P \dot{P}}}{\sin\alpha}$$

Now, we know,

$$0 \leq \sin\alpha \leq 1,$$

$$\frac{1}{\sin\alpha} \geq 1$$

So,

$$B \geq \left(\frac{3c^3 I}{8\pi^2 r^6} \right)^{1/2} \sqrt{P\dot{P}}$$

which represents the minimum possible magnetic field.

When we plug in the values for a typical neutron star, we get the minimum magnetic field strength at the surface of a canonical pulsar,

$$B \geq 3.2 \times 10^{19} \sqrt{P\dot{P}}$$

(in CGS unit gauss)

For the **Crab pulsar**, the minimum magnetic strength is

$$\left(\frac{B}{\text{gauss}} \right) \geq 4 \times 10^{12} \quad (1.12)$$

Characteristic Age:

Since $P_{rad} = -\dot{E}$, we go further and make assumptions about the Pulsar's age τ , subject to one condition. The pulsar's initial period P_o is much shorter than its current period P (we have approximated $(P^2 - P_o^2 = P^2)$ in the proof).

$$\tau \equiv \frac{P}{2\dot{P}} \quad (1.13)$$

For the Crab Pulsar, $\tau \approx 1240$ yrs, and it will continue to spin for millions of years.

Note that the equation (1.13) works on the approximation about the periods of the pulsars. As of 2026, the actual age of the Crab Pulsar is 972 years (since its supernova was observed in 1054 AD).

1.4 Braking Index

The braking index describes how the rotation of a pulsar slows down with time. It defines the proportionality relation between the rate of change of angular rotation frequency and the angular rotation frequency itself, ie.

$$\begin{aligned} \dot{\omega} &\propto -\omega^n \\ \dot{\omega} &= -K\omega^n \end{aligned} \quad (1.14)$$

where n here is the Braking Index.

Physically, n here is the opposing torque, that is slowing down the pulsar. We can very well see how the **braking index is closely linked with the Spin-down luminosity**.

For the Magnetic dipole radiation, the braking index is expected to be ≈ 3 .

$$\dot{\omega} = -K\omega^3$$

Now take the equation (1.14)

Differentiate both sides wrt time.

$$\ddot{\omega} = -K \left(n \omega^{n-1} \right) \dot{\omega}$$

$$\ddot{\omega} = -K \left(n \frac{\omega^n}{\omega} \right) \dot{\omega}$$

$$\ddot{\omega} = n \frac{\dot{\omega}^2}{\omega}$$

Rearranging,

$$n = \frac{\omega \ddot{\omega}}{\dot{\omega}^2} \quad (1.15)$$

The observed braking index values for pulsars are $1.4 \leq n < 3$.

1.5 The Life of a Pulsar

Let's look at the $P - \dot{P}$ Diagram, which is a visual representation of the entire life journey of the pulsar.

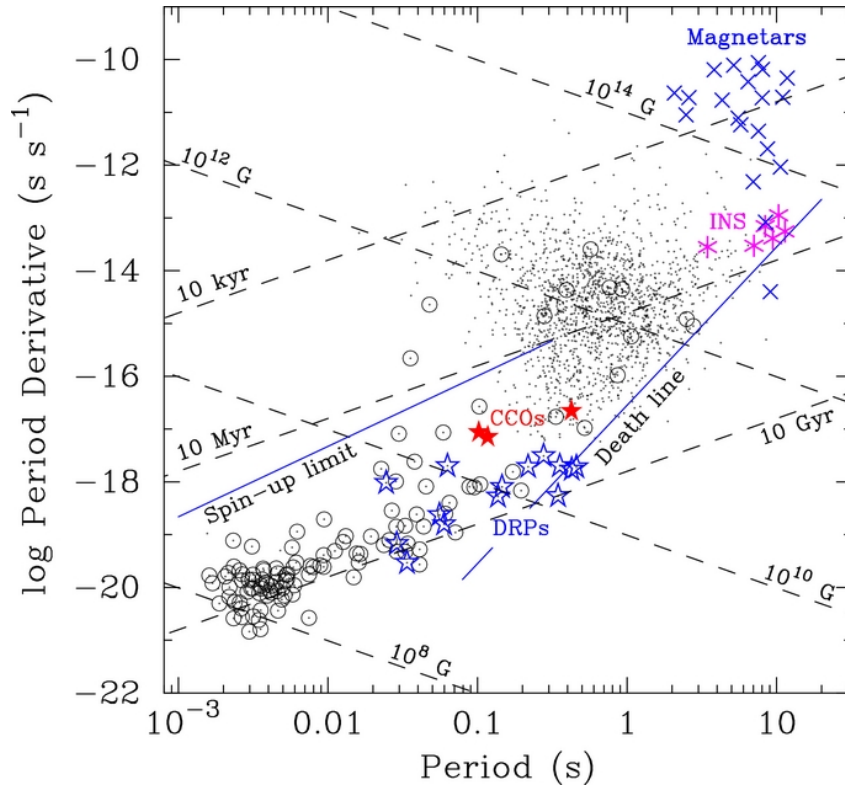


Figure 1.2: $P - \dot{P}$ Diagram of Pulsars

The spin period vs spin period derivative (how quickly the pulsars spin rate is slowing due to loss of luminous energy) diagram shows the different classes of neutron stars. From it, we have understood different properties of the neutrons stars, how they change over time, etc.

The life of pulsars is shown by the dotted line (with negative slope) in the above figure, starting at the top left corner and going towards the bottom right corner as its age goes on increasing. We can see the actual values as this happens (in CGS units Gauss).

The linear graph shows that as the characteristic age increases, the period of rotation also increases. It is understandable - a newly born neutron star has a very high angular momentum, so the pulsar 'pulsates' very fast. As the time goes on, the period of 'pulsation' goes up.

Also, the rate of increase of period (shown on the y-axis) is negative, which accounts for the energy reducing as the characteristic age of the star increases.

As the core collapses, electrons start combining with protons, forming neutrons and neutrinos, as seen previously in eq. (1.1). The density of the core increases rapidly, until it reaches that of a atomic nuclei. At this point, the core rebounds ie. produces a shock wave moving outward through the star. This is nothing but a **Supernova explosion**. What is left in the center is the neutron star (given it does not form a black hole), which is often observable due to the pulsars.

Interestingly, the pulsars with a characteristic age less than certain limit stay in the supernova remnants itself.

Rotating Radio Transients (RRATs):

Pulses from a pulsar are supposed to be very uniform. Their applications involve using it as clocks. But then there comes this species of pulsars called RRAT's. These, in a way, pulsate randomly.

These pulsars are observed to have $P > 1$ second, which means that it will lie in the right hand side of the graph. It is thought that these might be very old pulsars, almost close to turning off.

Recycled Pulsars:

Sometimes the binary system survives the supernova. In that case the neutron star keeps orbiting another star (the companion). The companion can be a normal star, or another neutron star. Now say, the magnetic field is very weak for the neutron star. It will get matter from the companion, and hence would have enough energy to emit radiation, forming a pulsar.

These are **millisecond pulsars**, meaning pulsars with period in milliseconds. These were not millisecond pulsars when they were born, but rather normal pulsars spun up by the accretion of the binary companion. These are rotationally more stable, and hence can act as good clocks.

1.6 Emission Mechanisms

Near the pulsar, there is a region filled with

- strong magnetic fields
- electrons and positrons (charged particles)
- intense electromagnetic radiation

This region is called **Magnetosphere**. The radio pulses originate in the pulsar magnetosphere. The neutron star is a dipole, whose magnetic axis is tilted at an angle to the rotation axis (it very well acts like a bar magnet spinning along an axis tilted to a small angle).

As the magnetic fields rotate, they induce an electric field. This is in accordance with Faraday's law. The neutron star acts as a **unipolar generator**, i.e., a conducting object that produces an electric field. Analogous to the bar magnet, the neutron star has two

poles at opposing ends. This generates a potential difference (voltage) between different parts.

The total **Lorentz force** acting on a charged particle is given by

$$\vec{F} = q \left(\vec{E} + \frac{\vec{v} \times \vec{B}}{c} \right) \quad (1.16)$$

Near the equatorial region, the electric charges redistribute themselves such that the electric field cancels the magnetic field, so $\vec{F} = 0$. The magnetic field lines emerging from the polar caps are supposed to close. However, the ones that cross the **light cylinder** (the distance where the corotation speed would equal the speed of light, i.e. the distance where the magnetosphere extends) do not form a closed loop as expected. This is because then the outer part of the line would have to travel faster than the speed of light, which is physically not possible.

Electrons in the polar cap region are accelerated to very high energies along the closed magnetic field (Any accelerating charge emits electromagnetic radiation. Even if the speed stays constant, a particle is accelerating if its direction changes.). These get accelerated to such relativistic velocities that they start emitting gamma rays, strong enough to interact with magnetic field lines and start producing electron-positron pairs. Now these pairs again interact with magnetic fields, and produce new gamma rays. And these gamma rays again interact to produce new electron-positron pairs, and so on. This makes them emit **curvature radiation** that is strongly polarised in the plane of curvature.

The pulsars 'die' as they lose enough magnetic field strength, and start having much higher time peroids. Graphically, the death is represented at the bottom right corner.

Chapter 2

Pulsars and the Interstellar Medium

A pulsar is a highly magnetized rotating Neutron Star. We get the radiations from these objects in the form of radio waves. The radio waves travel through the interstellar medium (ISM), which is ionized (contains weak magnetic fields). Radio waves are electromagnetic (EM) waves. In the influence of the weak magnetic fields, its plane of polarization will rotate, something known as **Faraday's effect**.

The refractive index for electromagnetic waves in a plasma is

$$\mu = \sqrt{1 - \frac{\nu_p^2}{\nu^2}} \quad (2.1)$$

μ is the refractive index of the EM wave

ν is the frequency of the radio wave

ν_p is the plasma frequency ($\sim 1.5 kHz$ for a typical ISM)

A special case arises when $\nu_p > \nu$. In that case the term inside the square root becomes negative, i.e. μ becomes imaginary. So, it is said that the radio wave simply cannot travel through the plasma.

For propagating radio waves, $\mu < 1$. The **Group Velocity** i.e. the collection of many waves with slightly different frequencies is given by

$$v_g = \mu c$$

For most radio observations, $\nu \gg \nu_p$. So we can apply binomial expansion on (2.1), and expand the term for group velocity as

$$v_g \approx c \left(1 - \frac{\nu_p^2}{2\nu^2} \right) \quad (2.2)$$

2.1 Dispersion delay

Dispersion delay is the extra travel time caused by the plasma comparison with travel in a vacuum.

The **Radio waves** are a collection of beams that slightly vary in frequency. The ISM contains free electrons. So there is a difference in the time required by the high-frequency waves (faster) and low-frequency waves (slower).

This time difference is called dispersion delay.

Time taken by the wave to travel small distance ' dl ' in plasma:

$$dt = \frac{dl}{v_g}$$

where v_g is the group velocity.

$$t = \int_0^d \frac{dl}{v_g}$$

Travel time in vacuum:

$$t_{vac} = \int_0^d \frac{dl}{c} = \frac{d}{c}$$

Dispersion delay is given by:

$$\int_0^d \frac{dl}{v_g} - \int_0^d \frac{dl}{c}$$

$$t = \int_0^d \left(\frac{1}{v_g} - \frac{1}{c} \right) dl$$

Substitute value for v_g from equation (2.2),

$$t = \frac{1}{c} \int_0^d \left(1 + \frac{\nu_p^2}{2\nu^2} \right) dl - d \quad (2.3)$$

Substitute the values of the constants,

$$\left(\frac{t}{sec} \right) = 4.149 \times 10^3 \left(\frac{DM}{pc.cm^{-3}} \right) \left(\frac{1}{(\nu/MHz)^2} \right) \quad (2.4)$$

where DM (Dispersion measure) is the total number of free electrons along the line the pulsar is being observed.

It is the total number of electrons that the radio wave has to travel through to reach the observer.

$$DM = \int_0^d n_e(l) dl$$

$n_e(l)$ is the electron density at a distance l (in electrons/ cm^3)

dl is the small segment of distance (in parsec (pc))

d is the total distance between observer and pulsar (in parsec (pc))

Now let us try plotting a graph for frequency (in Hz) on the y-axis and pulse phase (rotation in form of turns, written as 0 to 1) on the x-axis.

We see from the graph that the bright track is slanted. It represents the frequency dependent delay. From equation (2.4), we clearly see $t = 1/\nu^2$.

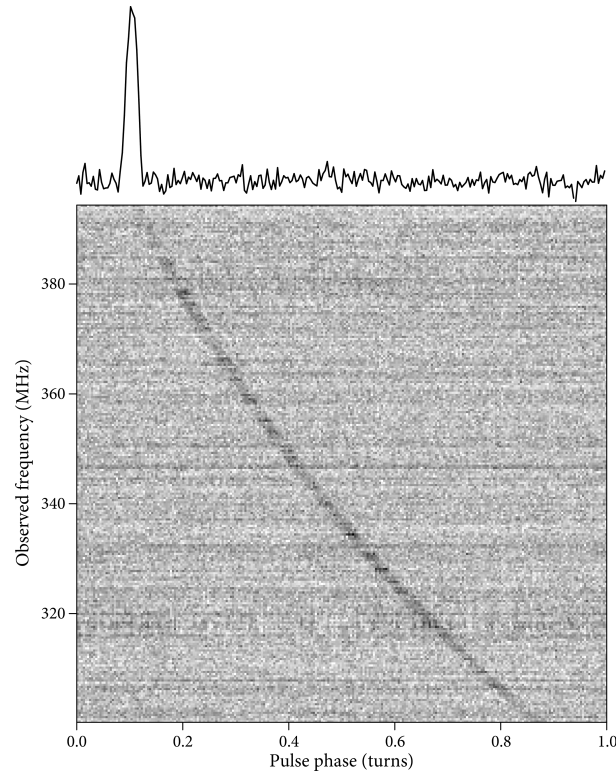


Figure 2.1: [Pulsar Dispersion](#)

In an ionized medium such as the ISM, lower frequencies travel slower than the higher ones. As a result, the pulse gets spread out in time, and is difficult to detect. This is called **Dispersive smearing**. Now, this is a problem. The best method useful for resolving such an issue is **Coherent dedispersion**. It works by multiplying by the inverse filter. This method removes delays as well as smearing

2.2 Scatter Broadening

A phenomenon related to scintillation (twinkling effect due to atmosphere) is **Pulse broadening due to scattering** of radio waves. Due to the ISM, different frequencies travel at different speeds. Some rays travel longer distances because they do not follow a straight line path.

From the figure that follows, we make the following observations:

- The pulse is sharp at high frequencies.
- At low frequencies, it becomes wide, asymmetric and develops a long tail.

It is said to reduce the observed flux density. Note that this does not mean that it destroys energy; it only redistributes it in time.

It becomes more difficult to determine the exact peak position. Also, it makes it difficult to obtain the true time of arrival of the pulse. The time is displayed between two vertical dotted lines in each scatter plot. Note how it increases and becomes more pronounced uncertain.

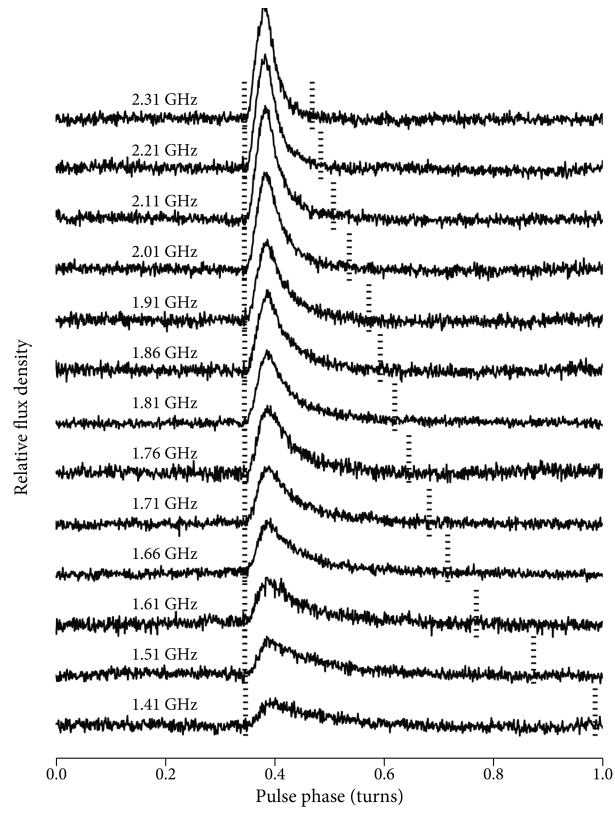


Figure 2.2: [Pulse broadening caused by scattering](#)

Chapter 3

Pulsar Timing

Now we move on towards the most interesting part - the actual use, why one should study pulsars. For pulsar timing, astronomers fold (take average) the data of the instantaneous pulse period P . The instantaneous pulse is of frequency $f = 1/P$, instantaneous pulse phase $\phi = \int f dt$ ($0 < \phi < 1$).

Averaging over many pulses gives an **Average pulse profile**. This adjusts noise.

We define **offset** as a measurable shift between two things. Here, phase offset is the measure of how much the observed pulse is shifted relative to template (ideal pulse profile).

Basically, $\Delta t = \text{Average pulse profile} \times \text{Instantaneous pulse period}$.

Time offset is considered to create **Time of arrival**.

$$\Delta t = \Delta \phi \times P$$

Time of arrival is this time offset added with the reference time.

(TOA is absolute time when the pulse arrives. Time offset is the relative shift)

3.1 Precision in Pulsar Timing

The precision with which TOA is measured (σ) is inversely proportional with **the signal to noise** (S/N) ratio of averaging profile.

$$\begin{aligned}\sigma_{TOA} &\propto \frac{W}{(S/N)} \\ \sigma_{TOA} &\approx \frac{W}{(S/N)}\end{aligned}\tag{3.1}$$

since proportionality constant ~ 1 .

where W is the effective pulse width. Smaller the value of W , sharper and narrower the pulse; and better the precision.

With what precision can timing determine the spin frequency f of a pulsar? The precision is based on how precisely a change in phase $\Delta \phi$ can be measured over some time interval ΔT from the relation $\Delta f = \frac{\Delta \phi}{\Delta T}$.

Now we take an example of the original millisecond pulsar B1937 + 21.

The time period P for the taken example is measured to be ≈ 0.00156 s.

Calculating,

$$\Delta f = \frac{\Delta \phi}{\Delta T} = \frac{6 \times 10^{-4}}{7.5 \times 10^8 \text{ s}^{-1}} = 8 \times 10^{-13} \text{ Hz}$$

$$f = \frac{1}{P} = \frac{1}{0.00156 \text{ s}} \approx 641 \text{ Hz}$$

So the absolute error is:

$$\frac{\Delta f}{f} = \frac{8 \times 10^{-13} \text{ Hz}}{641 \text{ Hz}} = 1.248 \times 10^{-15} \quad (3.2)$$

which is known to 15 significant figures!

Many **Correction factors** have to be checked before we proceed with our study of Pulsar timing. Precisely, the following equation takes all these factors into account.

$$t = t_t - t_o + \Delta_{clock} - \Delta_{DM} + \Delta_{R\odot} + \Delta_{E\odot} + \Delta_{S\odot} + \Delta_R + \Delta_E + \Delta_S \quad (3.3)$$

t_t (**topocentric time**) is the time at which the pulse is received at the telescope on Earth.

t_o (**reference time**) is a fixed reference time chosen as a starting point. It is an arbitrary point; but is fixed in the start.

So essentially,

$$t = t_t - t_o + \text{corrections}$$

Let us look at each of these delays in detail:

- **Clock correction** (Δ_{clock}):
The telescopes are not perfect. This correction is required to correct the differences between the observatory clock and the standard (say UTC time).
- **Correction due to Dispersion medium** (Δ_{DM}):
The Radio waves travel through the ISM. We have previously seen the effects this causes. This needs to be corrected.
- **Roemer delay** ($\Delta_{R\odot}$ and Δ_R):
The earth orbits the sun. The relative distance between the pulsar and the instantaneous position of earth changes. This accounts for the Roemer delay ($\Delta_{R\odot}$). Similarly, consider a binary system of neutron stars rotating along their center of mass. A similar effect will be noticed in this case too (Δ_R).
- **Einstein delay** ($\Delta_{E\odot}$ and Δ_E):
According to Einstein's special relativity, phenomena such as time dilation near massive objects are observed. The Sun's gravity causes redshifting as Earth revolves around it. Clocks run differently around these massive objects. Thus, there is a need for a correction factor (accounts for $\Delta_{E\odot}$).
Similar effects are visible in a binary star system as well (accounts for Δ_E)
- **Shapiro delay** ($\Delta_{S\odot}$ and Δ_S):
The Shapiro delay ($\Delta_{S\odot}$) is a result of curved space-time. The path near the massive is curved as a result of the gravity. So the it takes longer time to move along it. A similar delay (Δ_S) is observed when the space-time is curved along the binary star system. Again, it is same as Solar System Shapiro delay ($\Delta_{S\odot}$), but inside the binary.

To conclude, the errors are caused as a result of imperfections in the instrumentation, radio-wave propagation through the ISM, geometry (inside the solar system, or inside the binary), and relativity (inside the solar system, or inside the binary).

The binary pulsar Roemer delays comprise up to five **Keplerian parameters** describing elliptical orbits:

- the projected semimajor axis $x = (a \sin i)/c$
- longitude of periastron (ω) (the angle between a reference point and the point where the distance between the two stars of the binary are the closest.)
- time of periastron passage (T_0) (exact time when the binary system has the stars at least distance.)
- orbital period (P_b)
- orbital eccentricity (e)

Relativistic binaries, particularly those with compact and elliptical orbits, may allow the measurement of up to five **post-Keplerian (PK) parameters**.

Table 1 | Physical parameters for PSR J1614-2230

Parameter	Value
Ecliptic longitude (λ)	245.78827556(5) $^\circ$
Ecliptic latitude (β)	$-1.256744(2)^\circ$
Proper motion in λ	9.79(7) mas yr $^{-1}$
Proper motion in β	$-30(3)$ mas yr $^{-1}$
Parallax	0.5(6) mas
Pulsar spin period	3.1508076534271(6) ms
Period derivative	$9.6216(9) \times 10^{-21}$ s s $^{-1}$
Reference epoch (MJD)	53,600
Dispersion measure*	34.4865 pc cm $^{-3}$
Orbital period	8.6866194196(2) d
Projected semimajor axis	11.2911975(2) light s
First Laplace parameter (esin ω)	$1.1(3) \times 10^{-7}$
Second Laplace parameter (ecos ω)	$-1.29(3) \times 10^{-6}$
Companion mass	0.500(6) $M_\odot$
Sine of inclination angle	0.999894(5)
Epoch of ascending node (MJD)	52,331.1701098(3)
Span of timing data (MJD)	52,469–55,330
Number of TOAs†	2,206 (454, 1,752)
Root mean squared TOA residual	1.1 μ s
Right ascension (J2000)	16 h 14 min 36.5051(5) s
Declination (J2000)	$-22^\circ 30' 31.081(7)''$
Orbital eccentricity (e)	$1.30(4) \times 10^{-6}$
Inclination angle	$89.17(2)^\circ$
Pulsar mass	1.97(4) $M_\odot$
Dispersion-derived distance‡	1.2 kpc
Parallax distance	>0.9 kpc
Surface magnetic field	1.8×10^8 G
Characteristic age	5.2 Gyr
Spin-down luminosity	1.2×10^{34} erg s $^{-1}$

Figure 3.1: Physical parameters for PSR J1614-2239 (The numbers in parentheses for each parameter are the errors in the last digits of those parameters)

3.2 Pulsar Timing Arrays (PTAs)

The double neutron-star systems have indirectly confirmed the existence of gravitational waves by matching the predicted orbital decays as orbital energy is carried away by gravitational radiation. The key focus of pulsar astronomy has shifted towards the **detection of Gravitational Waves using Pulsar Timing Arrays (PTAs)**.

We have previously seen how stable the Millisecond Pulsars (MSP's) are. They act like precise clocks in space. Normally, an array refers to a collection of several telescopes that work together. But the PTA technique focused on a collection of multiple pulsars (rather than multiple instruments).

How does the technique work?

The time of arrival for a pulsar can be calculated theoretically. But the actual time varies. The gap is called **timing residual**.

$$\text{Residual} = \text{Observed} - \text{Predicted}$$

The most likely sources of detectable nanohertz GWs are supermassive black hole binaries (with total masses of $10^8 - 10^{10} M_{\odot}$).

In General Relativity, the gravitational waves are produced by **quadrupole changes** in mass distribution. By quadrupolar, we mean that the space is stretched in one direction and compressed in the perpendicular direction. Each pulsar sends regular pulses. In the presence of the GW's, there is a shift in pulse arrival time, timing residual.

If two pulsars are close together, the GW stretches space similarly along both lines of sight, their timing shifts are similar. This corresponds to positive correlation.

If pulsars are far apart, the GW stretches the space in one direction, and compresses in the other. Their timing shifts can be uneven (zero correlation), or even opposite (negative correlation).

Say we take a collection of pulsars. Then compare each pair, and plot the correlation value against the angle between them. What we get is the **Hellings–Downs curve**.

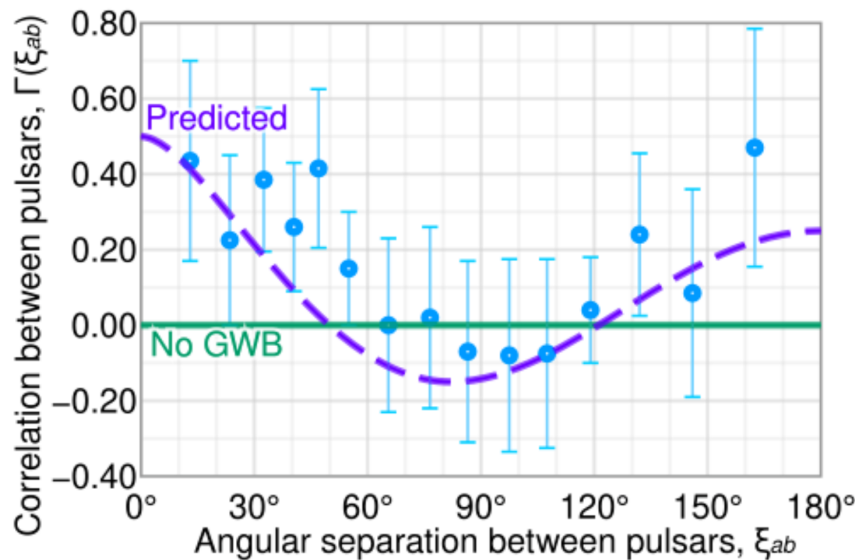


Figure 3.2: Hellings-Downs curve

3.3 Determination of Pulse Times of Arrivals

A paper by J. H. Taylor, titled "**Pulsar Timing and Relativistic Gravity**" (1992), provides a comprehensive review of pulsar timing techniques. TOA's are derived from averaging weak signals into profiles.

Let's say,

$p(t)$ = observed pulse signal

$s(t)$ = standard/template pulse

We sample (take measurements) at regular time intervals Δt .

So,

$$p_j = p(t_j); t_j = j\Delta t \quad (3.4)$$

$$p_0 = p(t_0); t_0 = 0$$

$$p_1 = p(t_1); t_1 = j\Delta t$$

$$p_2 = p(t_2); t_2 = 2j\Delta t$$

and so on.

Before sampling, we filter out the frequency range such that $f \leq \frac{1}{2\Delta T}$.
(where T is the emission time)

This is done to ensure that only the relevant signals are reached. To avoid filtering out useful data, Δt is taken very small.

If the above criteria are met, the **finite sampling theorem** ensures that the entire continuous signal is fully determined by its discrete samples. No information is lost. This is exactly the sampling theorem by **Nyquist-Shannon**; tailored for real, finite length data by Bracewell, 1965.

If $p(t)$ is equal to shifted and scaled replica of $s(t)$ plus some random noise, we can write

$$p(t) = a + b s(t - \tau) + noise$$

$p(t)$: observed pulse

$s(t)$: standard/template pulse

a : constant offset

b : scaling factor

τ : time shift

Importantly, a shift in time becomes a shift in phase in Fourier space. A shift in time changes the phase of frequency.

Specifically,

$$s(t - \tau) = e^{(i\omega t - i\omega\tau)} = e^{i\omega t} \cdot e^{-i\omega\tau}$$

where ω is a constant, and τ is the time offset.

The key identity is that

$$\text{Time shift} \implies \text{Multiply by } e^{-i\omega\tau}$$

It is preferable to deal in phase rather than in time (since phase can be treated as an **angle of a phasor diagram**). Take the Fourier transform of the observed signal. Then take the Fourier transform of the template. Compare phases (as angles of phasor).

We use the **Discrete Fourier Transform (DFT)** to convert the continuous real-world signals to discrete signals, which we are able to deal with.

Why?

Same reason, it is preferable to deal in phase rather than time. Discrete Fourier Transforms of the two profiles can be written as...

$$P_k \exp(i\theta_k) = \sum_{j=0}^{N-1} p_j \exp\left(i 2\pi j \frac{k}{N}\right) \quad (3.5)$$

$$S_k \exp(i\theta_k) = \sum_{j=0}^{N-1} s_j \exp\left(i 2\pi j \frac{k}{N}\right) \quad (3.6)$$

where P_k and S_k are the amplitudes of the complex Fourier coefficients (k is the frequency index), and θ_k and ϕ_k are the phases.

We write the combined equation as...

$$P_k \exp(i\theta_k) = aN + bS_k \exp[i(\phi_k + k\tau)] + G_k, \quad k = 0, 1, \dots, (N-1) \quad (3.7)$$

where the G_k term represents random noise. The bias a and scale factor b assume similar roles in the time and frequency domains.

After the transforms have been computed, the value of a can be directly obtained by putting $k = 0$, i.e.

$$a = \frac{P_0 - bS_0}{N}$$

The desired pulse time of arrival τ as well as the gain factor ' b ' can be obtained by goodness-of-fit statistics.

What is goodness-of-fit statistics?

It is the measure of how well the model matches the observer data. For example, here we try to match the observed pulse signal ($p(t)$) and the standard pulse ($s(t)$). The goodness-of-fit (χ^2) is given by:

$$\chi^2 = \sum \left(\frac{\text{difference}}{\text{uncertainty}} \right)^2 \quad \text{or} \quad \left(\frac{\text{noise}}{\text{uncertainty}} \right)^2 \quad (3.8)$$

Here,

$$\chi = \sum_{k=1}^{N/2} \left| \frac{P_k - bS_k \exp[i(\phi_k - \theta_k + k\tau)]}{\sigma_k} \right|^2$$

The additional term σ_k here is the root-mean-square amplitude of the noise at freq. k . This model tries to minimize this χ^2 . Smaller the χ^2 value, better the result. We try to get these a , b , τ such that the model best matches the data.

Lets understand why the limits are $k = 1$ to $N/2$. For that, we need to understand the symmetry.

For any real signal, $P_{N-k} = P_k^*$.

This is called conjugate/Hermitan symmetry (the $*$ represents the complex conjugate).

These are called the mirror copies of each other (same amplitude, opposite phase). The indices are equal in a way as:

$$k = N - k$$

$$1 = N - 1$$

$$2 = N - 2$$

(and so on...)

So we simplify the summation term.

$$\sum_{k=0}^{N-1} (...) = 2 \sum_{k=1}^{N/2} (...)$$

But the extra factor 2 is not required to be multiplied since each symmetric term represents both components.

From equation (3.7)

$$\begin{aligned} P_k e^{i\theta_k} &= b S_k e^{i(\phi_k + k\tau)} + G_k \\ P_k e^{i\theta_k} - b S_k e^{i(\phi_k + k\tau)} &= G_k \end{aligned}$$

This is now rearranged as data - model = noise.

From equation (3.8),

$$\chi^2 = \sum_k \frac{|P_k e^{i\theta_k} - b S_k e^{i(\phi_k - \theta_k + k\tau)}|^2}{\sigma_k^2}$$

But $e^{i\theta} = 1$, so

$$\chi^2 = \sum_k \frac{|P_k - b S_k e^{i(\phi_k - \theta_k + k\tau)}|^2}{\sigma_k^2} \quad (3.9)$$

Expand the Numerator:

$$\begin{aligned} |A - B|^2 &= (A - B)(A - B)^* \\ |P_k - b S_k e^{i\alpha_k}|^2 &= [P_k - b S_k e^{i\alpha_k}] [P_k - b S_k e^{-i\alpha_k}] \end{aligned}$$

(where $\alpha_k = \phi_k - \theta_k + k\tau$)

$$|P_k - b S_k e^{i\alpha_k}|^2 = P_k^2 - b P_k S_k e^{i\alpha_k} - b P_k S_k e^{-i\alpha_k} + b^2 S_k^2$$

But $e^{i\alpha_k} + e^{-i\alpha_k} = 2 \cos \alpha_k$

So,

$$|P_k - b S_k e^{i\alpha_k}|^2 = P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos \alpha_k$$

Substitute this back into eq (3.9)

$$\begin{aligned} \chi &= \sum_k \frac{P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos \alpha_k}{\sigma_k^2} \\ \chi &= \sum_k \frac{P_k^2 + b^2 S_k^2 - 2b P_k S_k \cos(\phi_k - \theta_k + k\tau)}{\sigma_k^2} \end{aligned} \quad (3.10)$$

A point to note here is that χ^2 is exactly the goodness-of-fit statistic here. The quantity does not precisely be in some range (like probability has a range 0 to 1). χ^2 **is a function of b and τ** . We do not care about the absolute χ^2 . We care about the value of τ that **minimises** χ^2 .

At the global minimum, the derivative of χ^2 wrt τ and b must be zero. Refer eq (3.10).

$$\frac{\partial \chi^2}{\partial \tau} = \frac{2b}{\sigma^2} \sum k P_k S_k \sin(\phi_k - \theta_k + k\tau) = 0 \quad (3.11)$$

$$\frac{\partial \chi^2}{\partial b} = \frac{2b}{\sigma^2} \sum S_k^2 - \frac{2}{\sigma^2} \sum P_k S_k \cos(\phi_k - \theta_k + k\tau) = 0 \quad (3.12)$$

Solving,

$$b = \frac{\sum P_k S_k \cos(\phi_k - \theta_k + k\tau)}{\sum S_k^2}$$

Any smooth function near a minimum behaves like a quadratic (parabola).

By Taylor expansions:

$$f(x) = f(x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2$$

For χ^2 :

$$\begin{aligned} \chi^2 &= \chi_{min}^2 + (curvature)(\tau - \tau_0^2) \\ \chi^2(\tau) &= \chi_{min}^2 + \frac{1}{2} \frac{\partial^2 \chi^2}{\partial \tau^2} (\tau - \tau_0^2) \end{aligned} \quad (3.13)$$

(This is the Taylor expansion around the best-fit value)

By definition of uncertainty (σ_τ), it is the shift in τ that increases χ^2 by 1.

$$\chi^2 = \chi_{min}^2 + 1$$

From eq (3.13)

$$\begin{aligned} 1 &= \frac{1}{2} \frac{\partial^2 \chi^2}{\partial \tau^2} (\sigma_\tau^2) \\ \sigma_\tau^2 &= 2 \left(\frac{\partial^2 \chi^2}{\partial \tau^2} \right)^{-1} \end{aligned}$$

The factor of 2 is "absorbed" because the summation is redefined to run only over independent frequency pairs.

So,

$$\begin{aligned} \sigma_\tau^2 &= \left(\frac{\partial^2 \chi^2}{\partial \tau^2} \right)^{-1} \\ \sigma_\tau^2 &= \frac{\sigma^2}{2b \sum k^2 P_k S_k \cos(\phi_k - \theta_k + k\tau)} \\ \sigma_b^2 &= \frac{\sigma}{2 \sum S_k^2} \end{aligned}$$

Chapter 4

Gravity and Gravitational Waves

4.1 Gravity Before and After Einstein

We are all much aware of the Newtonian definition of gravity, the model proposed by him and the famous **Newton's universal law of gravitation**, which states, "Every particle in the universe attracts every other particle with a force along a line joining them." This, however had its own flaws.

- What is physically carrying the two masses that are attracted, if gravity is a force? What is the pulling mechanism?
- Is this force instantaneous? If an object is moved, will the other object feel the effect immediately i.e. infinitely fast? This questions the theory of relativity that says nothing can travel faster than the speed of light.

Albert Einstein had already generalised Newtonian mechanics to arbitrary speeds $v \leq c$ in his 1905 breakthrough in special relativity, and went about doing the same for gravity. Einstein had a thought experiment which later came to be known as the [Equivalence Principle](#).

"Experiments in a sufficiently small freely-falling Laboratory, and carried out over a sufficiently short time, will give results that are identical to the same experiments carried out in empty space."

Minkowski developed an elegant geometric formalism for special relativity that recast the theory in a four-dimensional unification of space and time, known as spacetime. Even today, we refer to spatially flat four-dimensional spacetime as **Minkowski spacetime**. Note that Minkowski's work was on **Special Relativity**. In spacetime, an object moving at constant speed follows a straight path. If it is accelerating, its path becomes curved. Einstein realised that acceleration and gravity feel the same locally (this is the equivalence principle).

Special Relativity could not handle gravity. Einstein had found a correlation between acceleration and gravity. But SR works only in inertial frames. This was a problem!

The Happiest Thought:

The Newtonian way explains that if an object is falling, gravity is the cause, free fall is the effect. An object is dropped, it accelerates downward. Now imagine jumping with it. Suddenly the object will seem to float alongside us. The thought goes this way: If we can make gravity “disappear” just by choosing a falling frame, is it really a force? Gravity can locally be removed by going into free fall.

If gravity is not a force, it must be **something built into the structure of spacetime**. Special Relativity was an incomplete theory, by design, to explain gravity. It assumes the spacetime to be flat. Once this spacetime is allowed to curve, gravity was being able to be conceptualized. This curve in spacetime is nothing but gravity. This led to the development of the **General Theory of Relativity**. According to GR, momentum and energy change the geometry of spacetime. This spacetime is visually represented with the help of **Spacetime fabric**.

Einstein’s new theory described how spacetime geometry could cause bodies to follow curved free-fall paths known as geodesics. In essence, gravity is not a “pull” caused by a force-field; it is the by-product of bodies travelling along free-fall paths in curved spacetime. The shortest possible path between two points remains a straight line (of sorts), but for the Earth or other planets, this straight-as-possible path within the curved geometry created by the Sun is one that carries it around in an orbit.

Einstein’s curved spacetime idea was very well established, and sits in the peak of scientific theories as one of the twin pillars of 20th century physics, alongside quantum theory.

Einstein proposed three tests of his theory:

- Mercury orbits the sun in an ellipse. The ellipse has a closest point to the Sun, called Perihelion. The ellipse slowly rotates, as a result the perihelion shifts slightly after each orbit. This rotation is called perihelion precession. Mercury’s orbit precesses because it is moving through curved spacetime.
- Higher the mass, more effect it does on the spacetime fabric. Light passing near a massive object (like the Sun) gets bent, as a result. Gravity affects even light.
- This test is based on gravitational redshifting of light. Light has energy ($E = h\nu$). Light moving towards Earth is continuously falling into gravitational field, hence gaining energy. Frequency increases, and the light ‘blueshifts’. The opposite phenomenon is seen occurring while light goes away from gravitational bending of spacetime (redshifting).

Many more experiments have since validated GR.

4.2 Gravitational Waves

According to Einstein’s GR principle, if masses accelerate in certain ways, they disturb the spacetime fabric. These disturbances are called gravitational waves (GW). Now, are these GW’s similar to electromagnetic (EM) waves? The answer is no.

These waves behave much differently as compared to the EM waves.

The EM waves come from the dipole moment that is created between the positive and the negative charges present in nature. When it comes to the GW's, we don't have the concept of 'negative mass' in real life. The GW's hence are not the result of dipoles. We go further for them. The next more complicated structure is the **quadrupole moment**. A quadrupole describes how stretched or distorted a mass distribution is. This change in stretching or distortion of shape is what produces these GW's.

Specifically, we see that the second derivative of the quadrupole moment (we represent by Q) is proportional to the amplitude of radiation received by these waves.

$$h \propto \frac{d^2 Q}{dt^2}$$

A gravitational wave transports energy, momentum and angular momentum across space-time. They do work on objects, and hence transfer energy to matter. The real life proof to this comes from the energy profile of binary pulsars. We see that in the course of time, these tend to lose energy. This energy is emitted as the GW's.

The sticky bead argument by Richard Feymann:

This experiment imagined beads placed on a rod. When a gravitational wave passes, spacetime stretches and compresses, causing the beads to slide along the rod. Due to friction, heat is produced. Since heat requires energy, the energy must come from the gravitational wave itself, proving that gravitational waves are physically real and energy-carrying.

The 1970's saw the interests into detecting these GW's using **Laser Interferometry Techniques**. These early detectors had arm lengths from approximately 10-100 meters, and reached a strain sensitivity of approximately 10^{-18} for millisecond burst signals. Soon **Laser Interferometry Gravitational-wave Observatory (LIGO)** was setup at Hanford, USA and Livingston, USA. Other instruments have been built since, including the Virgo detector in Italy, and the KAGRA instrument in Japan.

4.3 Gravitational waves from General Relativity

General Relativity is enclosed within the Einstein Field Equations.

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

Conceptually, this says that matter and energy determine the curvature of spacetime. Then the gravitational wave is treated as a small oscillation in spacetime fabric. Assume that the spacetime is mostly flat, to make things easy. And now add a tiny disturbance to it.

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{4.1}$$

$\eta_{\mu\nu}$ is the perfectly flat spacetime.

$h_{\mu\nu}$ is the tiny disturbance.

Normally, this is much difficult to deal with. We take a few approximations, called the weak-field approximations. This is beneficial as Einstein's equations now simplify into wave equations. These waves are the Gravitational waves mathematically.

In gravitational waves, The quantity $h_{\mu\nu}$ describes tiny spacetime disturbances. But not every part of $h_{\mu\nu}$ corresponds to real physical deformation. This is where the Gauge freedom plays a role. Physicists remove the meaningless parts. Only two physically real components remain. These are the $+$ and $\times$ polarization.

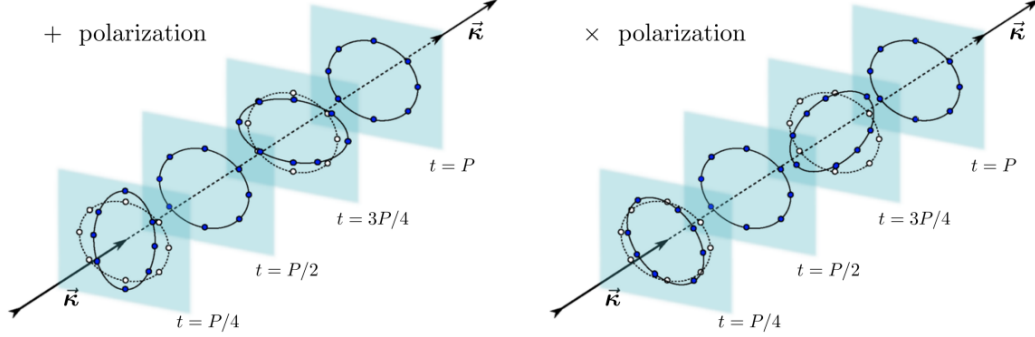


Figure 4.1: The periodic deformation of spacetime caused by a GW is illustrated for the two distinct polarizations permitted within GR. The influence of the GW is entirely in the plane perpendicular to the direction of propagation.

When a GW passes, space stretches in one direction, and compresses in the perpendicular direction. It is important to note that GW do not push everything in the direction it is propagating. It rather forces nearby space to deform in the directions of polarization. Nearby objects oscillate relative to each other. This is what detectors measure. The LIGO has two perpendicular arms, each arm being several kilometers long. A mirror is placed at the end of each arm.

Why are mirrors needed?

A gravitational wave changes spacetime distances very slightly. The mirrors move with spacetime because they are nearly free-floating.

When a GW passes, one arm becomes slightly longer, other one shorter. The laser travel times change slightly. This change creates an interference signal. The mirrors respond to the spacetime distortions.

We previously saw, that the first allowed radiation is the quadrupolar moment. Binary sources become an ideal GW source. Two orbiting objects produce a changing quadrupole moment. GW amplitude weakens with distance. The relation holding is

$$h \propto \frac{1}{r}$$

So the distance of the source can be approximated to the strength of the wave, which explains why GW detection is extremely difficult.

A detector may not individually resolve a number of systems emitting GWs at similar frequencies and comparable amplitudes. In such a scenario, the signals sum incoherently to produce a stochastic background of GWs (SGWB). By stochastic, we mean a random probability distribution that can be analysed statistically, but is almost impossible to predict.

4.4 Stochastic Gravitational Wave Backgrounds

We first talk about the energy density of a SGWB. The total energy of the universe is divided into mainly 3 categories - dark energy (68%), dark matter (27%), and the baryonic matter, i.e. everything else (5%). It is important to note that dark matter and dark energy are totally different things, and not at all connected.

The fractional energy density is usually defined in terms of closure density (ρ_c), defined as the density of the universe today that is required for flat spatial geometry.

$$\rho_c = \frac{3H_o^2}{8\pi}$$

where H_o is the Hubble's constant (70-76 kilometres per second per megaparsec). It is mathematically the expansion rate of the universe.

For a SGWB, it makes more sense to consider the fractional energy density as a spectrum, in order to see how this energy density is distributed over frequencies that may or may not be accessible to our detectors.

The energy density spectrum in a GW is defined as...

$$\Omega_{SGWB}(f) = \frac{1}{\rho_c} \frac{d\rho}{d \ln f} \quad (4.2)$$

where ρ is the energy density of gravitational waves. Physically, the higher the value of the fraction (Ω), the more energy there is at that frequency.

We write the gravitational wave as $h_{ab}(t, \vec{x})$. h here is a small perturbation or a small deviation in ab direction (ab are the coordinates). This is a tensor because a GW does not simply increase in all directions. It stretches a while, shrinking b and vice versa. A full tensor is required to describe the change in its geometry.

The entire background is decomposed by Fourier transform. The SGWB is treated as a superposition of many GW frequencies from many directions.

$$h_{ab}(t, \vec{x}) = \sum_{A=+, \times} \int_{-\infty}^{\infty} df \int_{S^2} d^2\Omega_{\hat{n}} h_A(f, \hat{n}) e^{2\pi i f(t + \hat{n} \cdot \vec{x})} e_{ab}^A(\hat{n}) \quad (4.3)$$

This Fourier transform includes the decomposition of the wave into the $+$ and $\times$ polarisation modes. Several assumptions are made about the background.

- **Gaussian:** Random fluctuations shall follow normal statistics.
- **Stationary:** Its statistical properties do not change with time.
- **Isotropic:** It is the same visually from all directions.
- **Homogeneous:** It is present the same everywhere statistically.

Solving the Fourier transform from (4.3), we obtain the following relation:

$$\Omega_{SGWB}(f) \equiv \frac{1}{\rho_c} \frac{d\rho}{d \ln f} = \frac{2\pi^2}{3H_o^2} f^3 S_h(f) \quad (4.4)$$

where $S_h(f)$ is the strain power spectral density of gravitational waves. This is the measure of how strong the gravitational-wave background is at each frequency. The h is the tiny strain as seen previously. It literally means a fractional change in distance ($h \sim \frac{\Delta L}{L}$). In the above relation, we see:

$$\Omega_{SGWB}(f) \propto f^3 S_h(f) \quad (4.5)$$

i.e., the energy density spectrum is directly related to the measurable strain power spectrum. This is a relation between the spacetime strain (h) and the energy density in GWs (Ω). This is important as the detectors measure the strain. But this strain is of no importance to us directly. What we are interested in is the physical energy density. This relation connects the two.

Characteristic Strain:

Similar to the final statement previously said, we are interested in characteristic strain, not the fractional energy density. It is represented as $h_c(f)$ and it is defined as:

$$h_c(f) = \sqrt{f S_h(f)}$$

Put that in equation (4.4),

$$\Omega_{SGWB}(f) \equiv \frac{1}{\rho_c} \frac{d\rho}{d \ln f} = \frac{2\pi^2}{3H_0^2} f^2 h_c^2(f) \quad (4.6)$$

4.5 Spectrum of the strain signal

We do not directly measure the metric perturbation ($h_{\mu\nu}$). We cannot do that because spacetime is not something we can see or physically feel. We rely on our detectors for this, which do not directly observe those abstract spacetime quantities. Instead, detectors measure physical changes (strain signals) produced by how the detector responds to the passing gravitational wave.

The data from the given detector is a combination of the measured signal ($h_i(t)$), and the noise ($n_i(t)$).

$$s_i(t) = h_i(t) + n_i(t)$$

where the measured strain signal is the projection of the GW metric perturbation onto the detector's response, such that

$$h_i(t) = d_i^{ab} h_{ab}(\vec{x}, t)$$

where d is the GW detector tensor of the i^{th} detector in ab space. A GW stretches spacetime differently in different directions. But a detector only measures along its arms (since that is where the mirrors are placed). So, physically, different detectors respond differently to the same GW. We take this time signal to frequency signal, because it gets easy to analyse by frequency.

For that we perform a Fourier transform.

$$\langle \tilde{h}_i(f) \tilde{h}_j^*(f') \rangle = \frac{1}{2} \delta(f - f') S_s(f)_{ij} \quad (4.7)$$

Let us understand the terms of this equation.

1. $\tilde{h}_i(f)$ means that we have done the Fourier transform on the perturbation, to express the GW as frequency.
2. $\tilde{h}_i^*(f)$ is the complex conjugate, standard to the Fourier analysis.
3. The angle brackets on the LHS are used to take the statistical average, since SGWB is random.
4. $S_s(f)$ is the Power spectral density (PSD), which measures how much signal power exists at frequency f in different detectors (here i and j represent different gravitational-wave detectors).
5. $\delta(f - f')$ is the Dirac delta function. This is the most important part of the equation. It acts as a perfect selector, such that $f = f' \implies$ that a correlation exists. While $f \neq f' \implies$ that no correlation exists. We make sure that,

$$\delta(f - f') = \begin{cases} \infty & f = f' \\ 0 & f \neq f' \end{cases}$$

Without the delta function, every frequency would correlate with every other frequency.

Now, similar to the electromagnetic waves, the gravitational waves also come in a spectrum of frequencies. The volume of such waves detected is huge, and it has not been possible yet to study all of those. There are three main detection schemes, though - the nanohertz scale (detected using pulsar timing arrays, as seen in previous chapters), millihertz scale (as seen in laser interferometry between space-borne satellites), and the 10-100 Hz scale (ground-based laser interferometry). We observe that as the scale gets smaller, the size of the detectors increases physically.

For megahertz scale for example, we will need even smaller instrumentation. On the other hand, for the detection of frequencies smaller than what PTA's could've detected, the size must be huge.

The reason why the space-borne detectors are in space is exactly this: the size. The millihertz scale detector for example, eLISA (evolving - Laser Interferometer Space Antenna) observatory will consist of three spacecraft arranged in an equilateral triangle with 2.5 million kilometres long arms. Having such a huge instrument on the planet would be impossible.

4.6 Sensitivity-Curve Plots of Gravitational Waves

In gravitational-wave astronomy, the three most common ways to represent GW strength on sensitivity plots are: characteristic strain (h_c), strain power spectral density (PSD) ($S_h(f)$), and the energy density spectrum ($\Omega_{GW}(f)$). We have discussed in detail the three previously also, but it is worth going through these once more (these are the three most common conventions we will plot against frequency: that is what we call a sensitivity-curve plot). We analyse them in the following order.

$$h \rightarrow S_h(f) \rightarrow h_c(f) \rightarrow \Omega_{GW}(f)$$

1. Characteristic strain:

This follows the very definition of strain. Strain (h) = $\Delta L/L$. Here, h is replaced by h_c as we see the L to be the arm length of the detectors. The larger its value, the stronger the detectable GW amplitude.

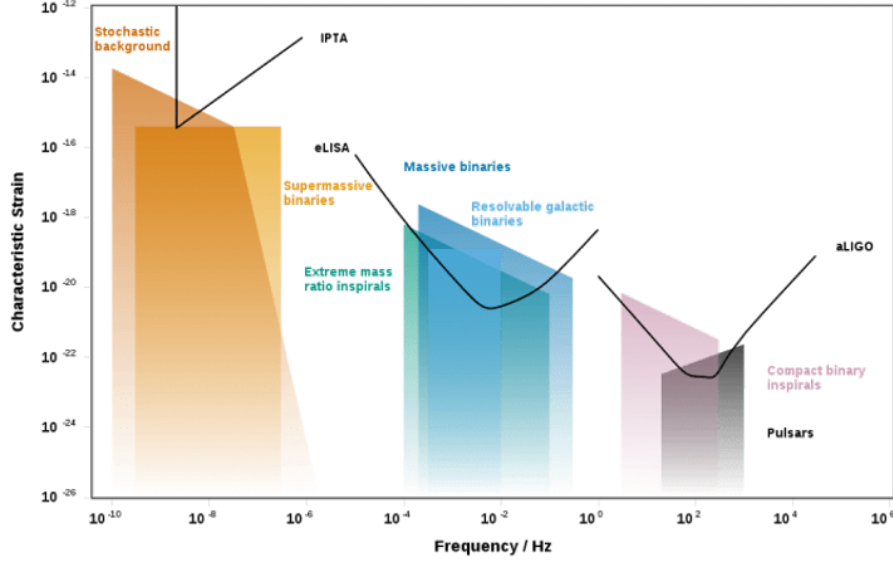


Figure 4.2: Gravitational-wave sensitivity-curve plot using characteristic strain. The area between the detector's curve and the top of the box for a source indicates how loud that signal would be.

2. Power Spectral Density (PSD):

$$h_c^2(f) = f S_h(f)$$

where S_h is the PSD. Also note that we are more interested in the Amplitude Spectral Density (ASD), which is the square root of PSD. Larger the PSD value, more is the GW power at that frequency.

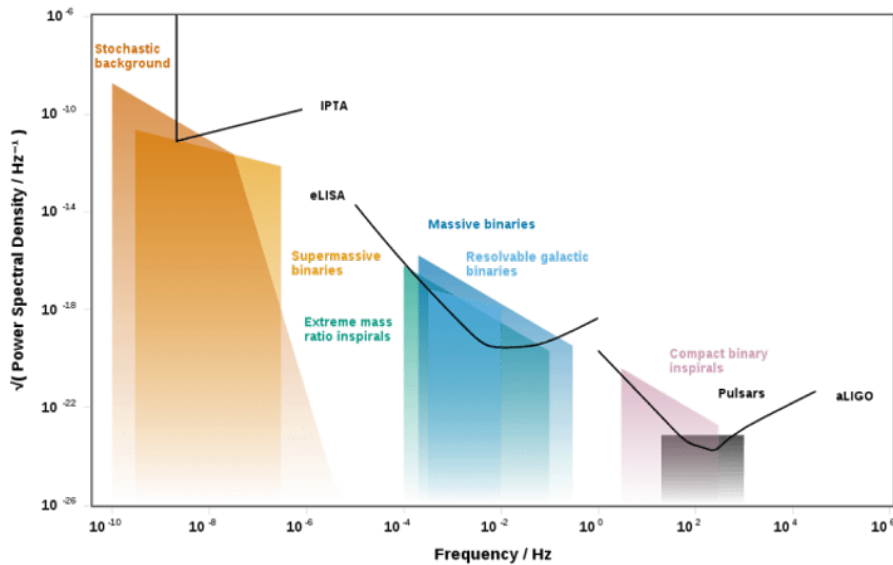


Figure 4.3: Gravitational-wave sensitivity-curve plot using the square root of the power spectral density (the amplitude spectral density)

3. Energy Density:

$$\Omega_{GW}(f) \propto f^2 h_c(f)$$

$$\Omega_{GW}(f) \propto f^3 S_h(f)$$

The larger the value of this parameter, the more energy is stored in GW's.

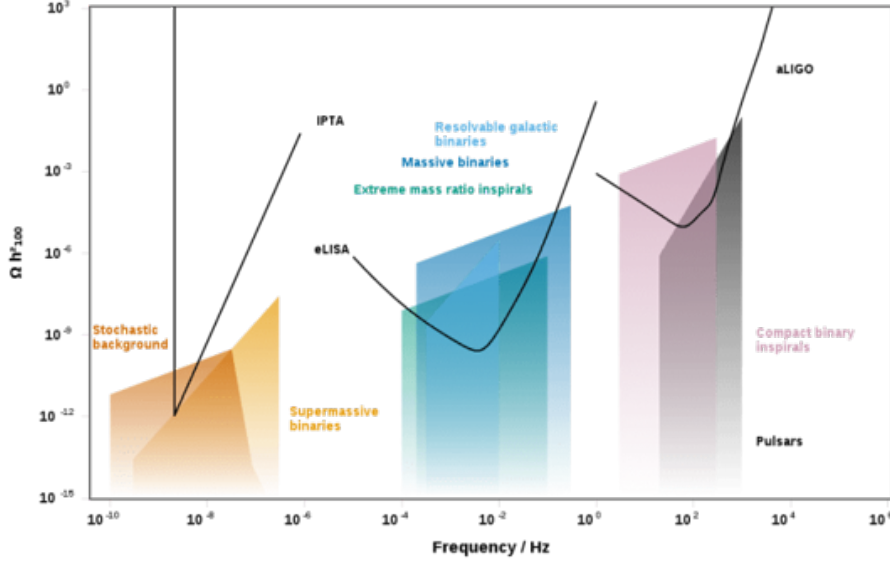


Figure 4.4: Gravitational-wave sensitivity-curve plot using the energy density.

4.7 GW Spectrum - Ground-based Detectors

The Weber bar (detector) was the first attempt towards the detection of GW's. It was a resonance detector, meaning that when a GW would pass through it, the detector would slightly stretch and compress. It had sensors that converted tiny mechanical vibrations into electrical signals. The narrow band sensitivity was the major problem for the Weber bar. Laser interferometry had the potential to overcome this limitation.

The working principle of the modern-day LIGO is laser phase interference. Inside LIGO, there are two perpendicular arms, present physically. These are 4 km arms, measured with great accuracy. A laser beam is split between the two arms, such that half of it goes to one arm, and the other half to the other arm. The beams reflect from the mirrors present at both arms and finally recombine at the initial site.

Now, a passing GW stretches one arm while compressing the other. The laser light thus contains some path difference. The waves may combine (constructive interference) or cancel (destructive interference). With LIGO, we are more interested in this path difference. This is much more accurate than the previous attempts. GW interferometers aim to achieve a strain sensitivity of $\sim 10^{-21}$ or lower.

Beyond these second-generation GW detectors, there are also now the third-generation ones in planning. These plan to achieve a broadband order of magnitude improvement in strain sensitivity and to push the operation down into the $1 - 10$ Hz frequency range. But one thing we understand is that as the frequency keeps dropping, we will be needing longer and longer detector arms. The calculations speak - after a certain limit (typically $0.1 - 100$ MHz), we cannot handle it terrestrially.

We thus decide to go to the space.

4.8 GW Spectrum - Space-borne Detectors

The canonical design for a mission in this band is the **Laser Interferometer Space Antenna (LISA)**. This project is led by the European Space Agency (ESA) in collaboration with the National Aeronautics and Space Administration (NASA).

This design integrates three identical satellites in a triangular manner, each separated by $2.5 \times 10^9 \text{ m}$. Now we understand why this was not possible on the planet. The test masses within each satellite are expected to be $\sim 46 \text{ mm}$, 2kg gold-coated cubes of gold/platinum (The choice of gold/platinum is since these metals are expected to float freely. Many materials are either magnetic, or they expand thermally, accumulate charge, or have other limitations such as these.) Each satellite will achieve zero-drag, which will allow it to absorb local gravitational influences. The LISA satellite constellation would trail the Earth's motion by $\sim 20^\circ$ as it orbits around the Sun as shown in the following figure.

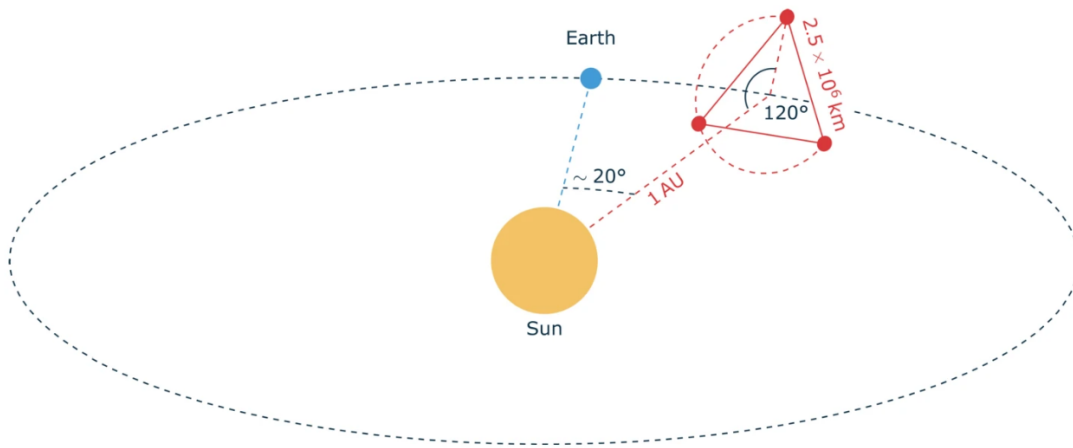


Figure 4.5: The LISA system trails the Earth by $\sim 20^\circ$ in the orbit

We call the path between the two satellites as **optical links**. There are three spacecraft - each pair exchanging laser beams. So, for each pair, there is a beam going one way, and one going the other way. In total, there are 6 optical links. Remember how the mirrors moved freely during the LIGO setup. The same task is taken up by the gold-platinum cubes that float freely. A passing GW changes the spacetime distances between these cubes. But unlike LIGO, the arms cannot be rigid here, since the distance is much greater. Each spacecraft sends a laser light to another spacecraft and receives the same from another. The phase difference (supposed to be zero in the absence of gravitational waves) is detected in the laser light, as a result of spacetime getting compressed and stretched at different distances. This is how LISA works.

The usage of an extra satellite instead of the usual 2 (as in the case of LIGO) is because of the cancellation of laser noise. Using the triangular geometry, beams travelling along different arms are combined mathematically, so the common laser fluctuations cancel out. This is the reason for counting the number of optical links as 6 and not 3. The geometry and timing of the laser paths allow these noise contributions to appear symmetrically, causing the cancellation. This is known as the Sagnac-type configuration. This technique of using symmetry to cancel out noise signal is called **Time-Delay Interferometry (TDI)**.

The prime target for LISA are the massive Black-hole (BH) binary systems of the range $10^4 - 10^7 M_\odot$. Detailed parameter estimation of individual systems will be possible, permitting inference of such effects like spin-orbit precession, higher waveform harmonics, and eccentricity.

A very interesting idea, such as the **multiband gravitational-wave astronomy**, is put forth. We are discussing about the detection of gravitational waves from a binary system, such as a black hole. As the binary system emits GW's, the energy and angular momentum is being carried away by these GW's. As a result of this, the orbit shrinks, and the BH's start approaching each other. This causes a change in the frequency of the GW's being emitted. As the objects spiral closer, orbital speed increases and so does the frequency.

The lower frequency can be detected by the LISA system, as it is sensitive to that. The higher frequencies can be detected by ground detectors like LIGO. These can detect only the final phase of the binary system and the merger. It will take $\sim$ months for the same wave to reach a frequency high enough for its detection by the ground-based detectors. But this idea effectively conveys the motivation to detect the same BH binary using both detectors within the appropriate frequency range. Another benefit it provides is that, even if LISA did not clearly detect a binary initially, a later detection by a ground detector can help us go back and find the earlier weak LISA signal hidden in old data.

LISA is an approved mission (officially adopted by ESA in 2024), and presently (as of 2026), in the engineering/construction phase. Its launch is planned for 2035, as of now. It is sure to bring in some exciting new insights towards the field of gravitational-wave astronomy.

Chapter 5

Sources and Signals

Now that we have enough idea about both Pulsar Timing Array (PTA) and Gravitational waves (GWs), we now try to connect these two topics. In this (and the following) chapter, we understand the various sources for PTA - the binary system of supermassive BH's, with masses $\sim 10^8 - 10^{10} M_\odot$. Beyond binaries, we understand a brief overview of other potential PTA sources, including primordial GWs, GWs from cosmic strings and phase transitions. Finally, we look over the dark matter that constitutes to $\sim 27\%$ of our universe.

Supermassive binary black holes:

This has become a general observation that most of the massive black holes (MBH) reside at the centre of a galaxy. So when we study any two black holes merging, we must consider the galaxies that get merged too. Indeed, these MBH's are among some of the loudest sources of PTA's which we are interested in. These are known as the Supermassive Black Hole binaries (SMBHB's). These now need to get very close in terms of separation - such that their GW emission would fall within the sensitive frequency range of PTAs ($\sim 1 - 100$ nHz). A chain of interactions happen, that lead to the BH's getting so close. Below a certain distance, the modern-day telescopes become incapable of resolving the two BH's.

5.1 Characteristic Strain Spectrum

We have seen in the previous section, eq. (4.4),

$$\Omega_{\text{SGWB}}(f) \equiv \frac{1}{\rho_c} \frac{dp}{d \ln f}$$

We write the RHS of the above equation as an integral over a continuous distribution of emitting sources:

$$\frac{dp}{d \ln f} = \int_0^\infty dz \frac{dn}{dz} \frac{1}{(1+z)} \frac{dE}{d \ln f_r} \bigg|_{f_r=f(1+z)} \quad (5.1)$$

where dn/dz is the number density distribution of binaries over redshift, $1/(1+z)$ is the cosmological redshifting of GW energy, $\frac{dE}{d \ln f_r}$ is the source-frame GW energy emitted by the binary per logarithmic frequency interval. The meaning of this equation is that: this is the integration of all the energies contributed from sources z to $z+1$.

Also, as seen previously,

$$h_c(f) \propto f^{-2/3} \quad (5.2)$$

We aim to understand how all SMBHBs collectively produce the gravitational-wave background (GWB) that pulsar timing arrays aim to detect.

$$\text{Total GW signal} = \int (\text{signal received from one binary}) \times (\text{total number of binaries})$$

The squared characteristic strain spectrum (we take the amplitude squared (h_c^2) instead of just the amplitude, because gravitational-wave power/energy scales as the square of the amplitude) can be written as

$$h_c^2(f) = \int_0^\infty \int_0^\infty \int_0^1 dz dM_1 dq \frac{d^4 N}{dz dM_1 dq dt_r} \times \sum_{n=1}^\infty \left\{ \frac{g[n, e(f_{K,r})]}{(n/2)^2} \frac{dt_r}{d \ln f_{K,r}} h^2(f_{K,r}) \right\} \quad (5.3)$$

where M_1 and M_2 are the masses of primary and secondary BH respectively. By primary, we simply refer to the one with a larger mass. 'q' is the ratio M_2/M_1 , $f_{K,r}$ is the source-frame Keplerian orbital frequency of the binary. The other terms of the integrand split into population factors and individual system factors. The function $g(n, e)$ is an eccentricity-dependent function describing the distribution of GW emission into orbital-frequency harmonics. It is important to know why this function is required. In circular binaries, the GW's are mainly emitted at one frequency. But for an eccentric orbit, the emission spreads into many harmonics. This function $g(n, e)$ tells us how much of the GW power goes into the n th harmonic for some eccentricity e .

For a circle, we put $e = 0$. The equation simplifies to:

$$h(f_{K,r}) = \sqrt{\frac{32}{5}} \frac{\mathcal{M}^{5/3} (2\pi f_{K,r})^{2/3}}{D_c} \quad (5.4)$$

where $\mathcal{M}$ (known as the chirp mass) $= \frac{(M_1 M_2)^{3/5}}{(M_1 + M_2)^{1/5}}$

and D_c is the radial comoving distance to the binary system. This coordinate system can be understood as the Euclidean system is not sufficient to deal with the universe's expansion.

And finally, the term $dt_r/d \ln f_{K,r}$ describes the amount of time that each SMBHB spends emitting in a particular logarithmic $f_{K,r}$ interval.

5.2 Binary Dynamical Evolution

The coalescence of two BH's is made possible in many steps, starting at merger of their host galaxies. From the equation (5.3),

$$h_c \propto \frac{dt_r}{f \ln f_{K,r}}$$

But

$$d \ln f = \frac{df}{f}$$

$$\frac{dt}{d \ln f} = f \frac{dt}{df} = \frac{f}{df/dt}$$

thus $dt/d \ln f$ is inversely proportional to the inspiral speed. If the orbit evolves slowly, the binary stays intact for a long time.

The question is whether GW's are the only responsible factor for the orbitals shrinking after the merger of two host galaxies? No. There are also the gravitational interactions with the surrounding matter that extract the energy and angular momentum from the binary. This increases the speed at which the binary evolves at low frequencies. As a result, the low-frequency GW signal weakens.

The factors that lead to the binaries evolving vary with the distance between the two BH's. When these are far apart, the GW emission is weak, so the surrounding gravitational interactions dominate. When these come closer, the GW emission takes over.

For binaries that evolve ONLY through GW emission,

$$h_c \propto f^{-2/3} \quad (5.5)$$

This physically means that as the frequency increases, the characteristic strain decreases slowly (we usually consider a log-log graph; here, it clearly decreases in a straight line with a slope of $-2/3$).

These environmental effects, however, create a *turnover*. The spectrum bends downward at low frequencies rather than following the normal behaviour. The new relation for h_c is given by

$$h_c(f) = A_{\text{SGWB}} \frac{(f/1 \text{ yr}^{-1})^{-2/3}}{[1 + (f_{\text{bend}}/f)^\kappa]^{1/2}} \quad (5.6)$$

where f_{bend} is the orbital frequency below which environmental interactions dominate the SMBHB evolution and A is the amplitude of the stochastic background. The denominator term is added to account for environmental effects.

The κ determines how sharply the turnover happens. The following parameters determine the value of κ .

1. Dynamical Friction:

Dynamical friction refers to the drag that the SMBHs experience while merging their galaxies. This results in the two systems getting closer, from kiloparsec separations to parsecs. It is thus an important step to initiate this reduction of distances.

2. Stellar loss-cone scattering:

At parsec-scale separations, the individual black holes are moving fast enough that dynamical friction no longer exerts a tightening drag. The binary gets closer, thanks to various factors, and becomes harder to break. The stars with low angular momentum thus scatter out, and what is left is called a loss-cone. A star approaches the binary and undergoes a gravitational slingshot interaction. As a result, the star gains kinetic energy and is flung out to a much larger orbit. This is considered to be removed from the **loss cone (LC)** (cone because in velocity space, these orbits form a cone-shaped region). But there is a problem. If the only stars present in the lost cone region go missing one by one, it will eventually empty. Then the hardening of the binary systems would tend to cease. This is known as the *Final Parsec Problem*.

Fortunately, galaxies are not perfectly static. The loss cone never practically empties.

If we consider the LC scattering as the dominant term in $dt_r/\ln f_{K,r}$, then the resulting strain spectrum behaviour is $h_c(f) \propto f$ and not the usual $\propto f^{-2/3}$ (circular orbit behaviour). This corresponds to $\kappa = 10/3$ in equation (5.6).

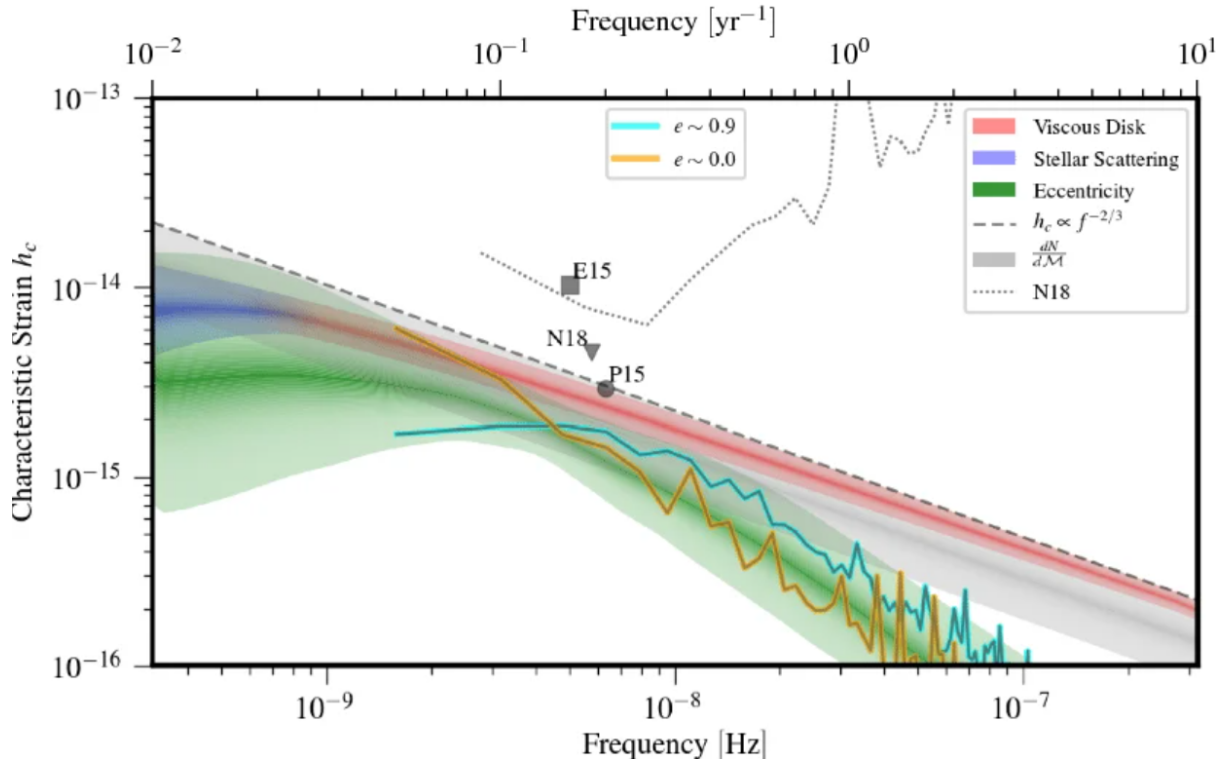


Figure 5.1: An example of the characteristic strain spectrum of a SGWB of SMBHB origin, as a function of GW frequency. The grey dashed line shows the $f^{-2/3}$ spectrum, while the influences of dynamical interactions are shown as colored bands, for stellar loss-cone scattering (blue) and viscous disk coupling (red).

3. Viscous circumbinary disk interaction

Now, let's take a look at centiparsec-to-milliparsec separations. A huge disk of gas surrounds the binary system. This, obviously, is not frictionless. Different parts of the disk move at different velocities; as a result, they rub against each other. Some energy and angular momentum is thus lost in the form of heat. The binary system has a strong gravitational torque (torque, because the system itself is rotating). This torque is much stronger than the viscous (or frictional) force internally acting in the gases. As a result, gas near the binary is pushed away, and a cavity is formed in the centre. This cavity is filled by the binary systems getting closer together. This type of in-filling of a space is known as **Type II migration**.

The binary's gravitational torque results in spiral patterns of the gas surrounding the binary. More and more gas accumulates as time passes, rotating at extreme speeds. These are high-density spiral waves, which carry angular momentum. Once these are projected and launched out from the system, the disk gains all the angular momentum, and the binary loses it all. Thus, the orbit shrinks - driving toward the merger.

The strain spectrum under different scenarios can vary from $h_c(f) \propto f^{-1/6}$ ($\kappa = 1$) to $h_c(f) \propto f^{1/2}$ ($\kappa = 7/3$). As we observe, the characteristic strain spectrum can be flattened or even have a positive slope due to disk coupling.

We are now heading to extremely small separations between the two individual black holes. The GW emission phenomena begin to dominate over the surrounding gravitational interactions.

4. Gravitational-wave inspiral

The separation now reaches an extremely small scale ($< \text{milli-parsec}$). The dissipation of orbital energy then only depends on the binary component masses, the orbital semi-major axis, and the eccentricity. We have already seen that this leads to $h_c \propto f^{-2/3}$. However, this is an averaged power law. As the frequency crosses 10^8 Hz , the slope will get steeper, and this relation will start getting violated in the reverse direction

A problem that may arise while the binaries come closer is what if they are unable? What if all the above 3 discussed mechanisms fail to reduce the separation to a scale where the merger is expected?

This is called **Binary stalling**. Another unique mechanism is put forth for the rescue of such a condition. Triplet interaction - add another BH in the system. The three-body problem is chaotic. But energy and angular momentum get exchanged very efficiently. Typically, the outcome of this is that one of the two initial BHs gets kicked out of the system, and the rest go on to merge. Galaxies can undergo numerous merger events over cosmic time, with the potential to form massive BH systems that undergo **Kozai-Lidov oscillations**. This is an event that occurs specifically when a third BH intervenes in a binary system (we are specifically interested in a stalled binary system). The gravitational influence of the third black hole causes perturbations in the binary systems, causing cycles of eccentricity going up and down, and changes in inclinations.

A strong GW background simply means that the evolution is slow, which means the binary spent more time in a particular frequency range we were interested in (as it did in every other frequency). Similarly, a weaker background arises when the binary moves quickly through that particular frequency bin.

5.3 Gravitational-wave memory burst

The final phase is the merger. This phase results in the emission of huge amounts of energy. The final merger frequency of these SMBHs lies in the range $1 - 100 \mu\text{Hz}$. PTAs observe the nano-hertz band while LISA observe the milli-hertz range. We take the indirect help of PTAs. The normal GW behaves as a distortion of the spacetime fabric - stretches and compresses and returns to normal periodically. After the merger, the spacetime is left slightly shifted instead of returning to the exact original level. There is a permanent offset known as **gravitational-wave memory**. After the merger, a burst of GWs is emitted. The burst leaves a permanent shift in the spacetime fabric.

The question is - how do we detect it? We know that PTAs detect the pulse time arrivals. After the merger, the displacement of spacetime would cause a slight change in the effective path length of the wave. The time period would not be affected in general, except for the exact moment the merger is detected (only that part of the wave would have a delay). The memory does not appear instantaneously. For a $10^9 M_\odot$ merger, the offset builds up over a day.

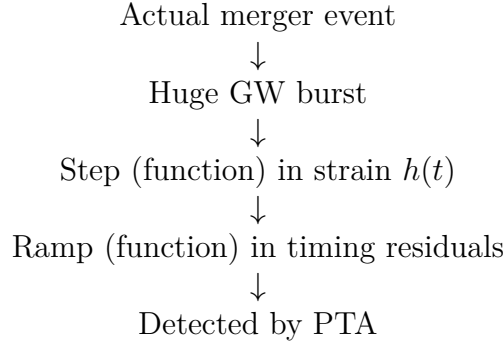
We can measure the GW memory burst as a step-function in strain. A normal gravitational wave strain oscillates. But as the memory burst event happens, there is a sudden change in the value of the characteristic strain (h_c). This is the step-function, that is the result of the permanent offset acquired by the spacetime fabric. We know that the PTAs measure timing residuals, which are changes in pulse arrival times.

Now roughly,

$$\text{timing residual} \propto \int h_c(t) dt$$

Integrating the step-function gives a ramp-function. A ramp means that the residual grows linearly (like a ramp) with a constant slope.

Now, PTAs do not observe the pulsars continuously. They usually observe a pulsar once a week or even months. The merger event usually takes a day, as previously stated. So, although it appears to be an immediate event (and the step function), it is not.



While SMBHBs contribute to the major sources of PTAs, there may be additional *exotic sources of GWs* lurking beneath the binary signals.

5.4 Relic/Primordial Gravitational waves

The electromagnetic observations, i.e., those detected using the light spectrum, are limited to a time scale in the past. We cannot use them to understand the universe from its very birth. A portion of it, from birth to about 3,80,000 years after that is missing. The oldest light we can directly observe is called the **cosmic microwave background (CMB)**. This is because in that time period, the Universe was filled with hot ionised plasma. Photons could not travel freely in space. This is where the discovery of Gravitational waves helps. We try to study gravitational waves generated extremely early in the Universe, possibly during inflation. Unlike those discussed previously, these GWs are not produced by the merger of two giant astrophysical objects. These exist due to the universe itself. So these can allow us to study the universe far before the CMB.

Gravitational waves interact extremely weakly with matter. This allows them to travel easily through the universe. Practically, a gravitational wave created shortly after the Big Bang could travel almost unchanged until today. The theory touches on the inflation that occurred during the initial phase of the Big Bang. The tiny quantum fluctuations evolved into giant cosmological level fluctuations. This results in the GWs emission.

5.5 Cosmological phase transitions

The initial phase of the universe was very hot. As it expanded and cooled, different states of matter (other than the initial plasma) began to appear. This is the result of the adiabatic expansion. As the universe cools, the states of matter begin to coexist. As these

are coexisting, and the temperature continues to drop, the newer state is now energetically preferable. Tiny regions of the newer state start to emerge in the space. These tiny regions are called bubbles. As these bubbles grow, they meet one another and eventually fill in the gap. In technical terms, the old state of the universe is termed the 'false vacuum'. As the transition to 'true vacuum' begins, the bubbles collide. These collisions shake the local spacetime (since each bubble carries enormous amounts of energy). This shaking of the spacetime fabric is what creates the gravitational waves.

These bubble collisions further give rise to another phenomenon. The sound waves that were generated within the false vacuum by the original bubble wall collisions collide with the plasma turbulence generated by the expansion and collision of the sound waves.

The spectrum of GWs generated by first-order phase transitions does not obey a power-law shape.

5.6 Cosmic strings

We now move towards a theoretically proposed source, accepted by various models. We have discussed that the early universe underwent phase transitions as it cooled. Different regions of the universe may have chosen slightly different 'true states' as they evolved. When these regions met, defects could have formed. One such defect is a **cosmic string**. It is a very thin region where the 'false state' remained trapped. These tiny segments could have enormous gravitational effects. These strings are proposed to form loops, cross each other, and reconnect. We are interested in the loops that get formed. As the loop wiggles, its shape, the quadrupolar moment, changes. As a result, it emits GWs. Together, many loops emit GWs; as a result, they create a stochastic gravitational-wave background (SGWB).

Another such GW emission theory has been proposed. As seen, many cosmic strings exist in the universe. This collection is called the cosmic string network. If we assume that these never emit out energy, we will eventually run into a problem of infinite energy saturation (since the universe is expanding, and these strings are gaining energy from the expansion). The loop loses energy as it emits gravitational waves. As a result, it shrinks. Its size decreases, and eventually it completely disappears. This is called a loop decay. However, a model is yet to predict a purely power-law spectral shape across the entire GW spectrum. But approximations are for $\alpha = -7/6$ and $\gamma = 16/3$.

There is also interest in searching for GW bursts from individual cosmic string cusps and kinks (sharp corners as the loops oscillate).

Now, further, we will look at a few sources of timing delays and inter-pulsar correlations that may originate from the Earth, the solar system, and the galaxy.

Chapter 6

Non-GW sources of correlated timing delays

Let us now return to the topic of PTAs, but not the one related to gravitational waves. Not every correlated signal is caused by gravitational waves. Some natural effects can make many pulsars appear to fluctuate together, even when there is no direct or indirect involvement of gravitational waves.

6.1 Clock errors

Atomic clocks are found all over the world. It is important to combine measurements from hundreds of atomic clocks. **BIPM or Bureau International des Poids et Mesures** (International Bureau of Weights and Measures) is responsible for this. It is also responsible for the maintenance of SI units, international seconds, and various such measures related issues. BIPM issues the **TAI (International Atomic Time)** - the world's fundamental atomic clock timescale. Astronomers need a very stable time scale for calculations. The new modern atomic clocks are very stable indeed. But PTA calculations are mostly done in $10^{-6} - 10^{-7}$ seconds, causing a problem in accuracy.

Consider a scenario - the clock got shifted by 10 ns. The pulsar just after the shift will record a shift of (expected time + 10 ns residual). The same will be observed in all the pulsars thereafter. Each gets a delay of 10 ns (Assuming that the atomic clock works perfectly fine thereafter). In such a case, it is important to note that there are two possibilities to this. Either there is a real gravitational wave that got recorded, or it is merely a clock error.

The real question is - how do we resolve this? To answer this, we have to look at the correlation, and how it changes with the changes happening in the angle between the two pulsars. If it is a clock error, then it would affect all the pulsars next to it in an equal manner (a correlation of 1). This is called **monopole**. While in the case of a gravitational wave, it stretches differently depending on the direction. The correlation is not constant. It follows the **Helling's-Down's curve**.

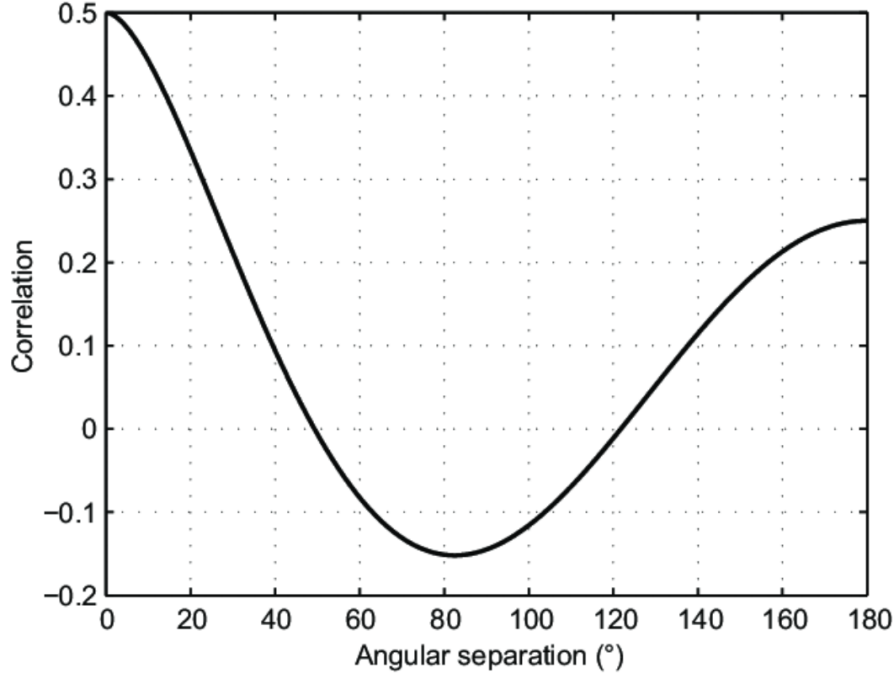


Figure 6.1: [Hellings-Downs curve](#)

6.2 Solar System ephemeris errors

The Hellings-Downs curve very well shows how pulsar timing residuals are obtained by subtracting the observed TOAs from a best-fit model. Fig 7.1 shows a smooth curve of what was a noisy collection of data. The observed TOA must be analysed to fit the time the event actually happened. This requires a chain of correction as seen in section 3.1. Another important correction that must be done is the uncertainty that is encountered while we sort out the frame of reference. When a pulse arrives at your telescope, you don't analyse it as if it arrived at Earth. Instead, it is assumed that the pulse has arrived at the Solar System Barycenter (SSB). But to compute this, we need to have an absolutely accurate model of the galaxy.

Ephemeris is the precise model/table that talks about the position and velocity of a body in space throughout its lifetime (future also). The Roemer delay deals with this.

$$\Delta R_{\odot} = t^{\text{obs}} - t^{\text{SSB}} = - \left[\vec{r}^{\text{obs}}(t^{\text{obs}}) - \vec{r}^{\text{SSB}}(t^{\text{obs}}) \right] \cdot \hat{p} \quad (6.1)$$

where $\vec{r}^{\text{SSB/obs}}$ are the coordinate vectors of the observatory and SSB at the observed pulse arrival time, respectively, and $\hat{p}$ is a unit vector pointing in the direction of the pulsar.

Any imperfections in the ephemeris will cause significant errors in the above equation. The process of analysing the pulse, as to when it should have been reaching the barycenter is known as Barycentering. Following equation describes the the timing error caused by the wrong Solar System ephemeris:

$$\delta t_{R_{\odot}} = -\delta \vec{x}^{(3)}(t^{\text{obs}}) \cdot \hat{p}$$

where the LHS of the equation is the error in timing, while the RHS is the error in position (as a function of time, in the direction of the pulsar denoted by $\hat{p}$).

The error in the position is caused as a result of the imperfect location of the barycenter of the solar system. It is not a fixed point in space. Instead it is the center of mass that takes into account every mass in the system. Given that the masses are not static, so is the position of the barycenter.

6.3 Dark Matter

There are two standard models associated with dark matter theory. The major difference comes in the mass of the two, and its composition.

1. Cold Dark Matter (CDM):

This is a more standard model among the two. CDM model talks about dark matter's gravitational influence on pulse arrival times. Dark matter does not form a smooth halo around our galaxy. Instead it consists of many tiny clumps, ranging in sizes. If a clump passes near to the earth (where detector is located) or the pulsar (the source of PTAs), the gravitational force of that clump shall affect the pulse time arrival. This creates time residuals. Now the position of this clump especially depends whether it is near the source or the detector. This can be called the **Doppler effect of pulse time arrival**.

If the clump is near the Earth, each pulse arrival time will get affected till the time the clump is in range of showing gravitational force.

If the clump is near the pulsar, only the first pulse time will create the residual. The rest will follow the expected time (+ residual of the first one).

There is another effect that plays a role. Lets say, the DM clump is between the pulsar and Earth. Its gravity obviously would not affect the pulses. However, it can slow down the pulse. The gravity of the DM clump would bend the spacetime fabric, to a small extent. That would mean that the pulse now has to travel a longer path in order to reach the Earth. This would create a delay, and in turn - time residual. This is what scientists call **Shapiro delay**.

2. Fuzzy Dark Matter (FDM):

This theory suggests that dark matter consists of extremely light particles, mass of the order $\sim 10^{-22} \text{ eV}$. At such low masses, the de Broglie wavelength becomes extremely large.

$$\lambda_{dB} = \frac{h}{mv}$$

For $m \sim 10^{-22} \text{ eV}$, the $\lambda_{dB} = 60 \text{ pc}$. This is an extremely large wavelength, comparable at astrophysical scales. It is called fuzzy, in the opposite sense to smooth.

Think of the DM clump as a wave, and not a particle.

$$\phi = \cos(mt)$$

The CMD theory stated that the dark matter clump induced a static gravitational potential. This, on the other hand suggests that the induced potential is oscillatory. Let us take a look at the frequency of this oscillation.

$$f = \frac{2mc^2}{h} = \frac{2m}{h}$$

(We take $c = 1$ for convenience. The units are taken care.)

Now plugging in $m = 10^{-22} \text{ eV}$, we get $f = 4.8 \times 10^{-8} \text{ Hz}$. Notice that this frequency lies in the microhertz frequency band, in which PTAs work.

The major difference between CMD and FMD is that in CMD, we observe random timing residuals, while in FMD, those are sinusoidal. This is why PTAs are not only gravitational-wave detectors. They are also potentially ultralight dark matter detectors.

Chapter 7

The Past, Present, and Future of PTAs

By the very nature of Pulsars, the PTA experiments are long-timescale projects. On a global scale, the **International Pulsar Timing Arrays (IPTA)** take charge of all the projects worldwide. It is important to know about IPTA before we proceed.

7.1 About the International Pulsar Timing Arrays

In the early 2000s, there existed three main PTAs -

- **European Pulsar Timing Array (EPTA)**, established in 2004 is a collaboration of PTA observatories from five European nations - France, Germany, Italy, the Netherlands, and the UK.
- **Parkes Pulsar Timing Array (PPTA)**, Australia, was established in 2004 using the Parkes Observatory.
- **North American Nanohertz Observatory for Gravitational Waves (NANOGrav)**, in the US. Established in 2007, it uses the Green Bank Telescope, and the Arecibo Observatory (until its collapse in 2020).

As scientists realised that different pulsars have been analysed by each of the three regional PTAs, there was the establishment of the International Pulsar Timing Arrays in 2009.

In 2016, the **Indian Pulsar Timing Arrays (InPTA)** was established, and became a member of the International PTA in 2021. Today, it works with the collaboration and efforts of these four regional PTAs. Aside from these four, efforts are also underway by the **Chinese Pulsar Timing Arrays (CPTA)**, despite not being a member of the IPTA (as of June 2026).

The IPTA has produced two data releases so far: DR1 (2016) and DR2 (2022). The first data release had data from a collection of 49 millisecond pulsars, taken in the last 30 years approximately. The second one had data from a collection of 65 millisecond pulsars. The main purpose was improved timing precision, better noise modelling, and, of course, more pulsars.

The ultimate aim remains the same - detecting the stochastic GW background.

7.2 The Future of Pulsar Timing Arrays

So what's next? Either of the PTA collaborations has to announce the detection of the Stochastic Gravitational Wave Background (SGWB). This must be confirmed (or refuted) by other regional PTAs and the IPTA. The first idea of the PTAs occurred in 1983, the same time as the Hellings-Downs curve was discovered. That paper essentially founded PTA gravitational-wave astronomy. The present precision should be sufficient to test some astrophysical models of the SGWB and discriminate the origin under simple assumptions of the spectral index. As we succeed in doing this, we can then try out different models, and measure the deviations from the expected $h_c(f) \propto f^{-2/3}$.

That's not it. Other future goals include...

- **Test theories of gravity beyond Einstein's**

Einstein's General Relativity (GR) predicts that gravitational waves have two polarisation modes (the + and the $\times$). An alternate theory would aim to find more polarisation patterns than these two. Also, the actual speed at which the gravitational waves travel is into discovery. The argument is that - if these are not the regular Electromagnetic waves, they need not necessarily follow the speed of light barrier. If they exceed that barrier, an entire section of physics can be questioned.

- **Search for primordial gravitational waves**

As discussed previously, the PTA and gravitational waves are helpful in the study of the early universe, even before the CMB. As this field develops, we can get closer to the actual Big Bang event.

- **Challenging the Big Bang theory itself!**

PTAs can also be a source to challenge the theory itself, as done by Padma Vibhushan Shri Jayant Narlikar in 2025. He brought in the Hoyle–Narlikar theory of gravitation. This was a big thing, as it led to the development of the Quasi-Steady State Cosmology, an alternative vision of a universe without a beginning.

- **Search for cosmic strings**

If they exist, they should continuously emit gravitational waves as they oscillate or reconnect. Detecting such a signal would provide evidence for physics at energy scales far beyond what particle accelerators can reach.

- **Look for evidence of first-order phase transitions**

Some theories predict these transitions happened violently, with bubbles of a new phase expanding and colliding. While some come up with different predictions. Those collisions would create a stochastic gravitational-wave background. Detecting this would help us confirm us that the standard model of particle physics (the extensions) is correct.

- **Search for Dark Matter**

This is a broad goal, and is expected to take a good amount of time as we reach here. If PTAs could detect gravitational-wave signals that don't fit existing models, then that could be the clusters of dark matter. This could discover an entirely new cosmological phenomenon.

The broader goal of these missions is the overall development of Gravitational-wave astrophysics. In the process, automatically, newer and better telescopes and other instrumentation will be built. Newer analysis, statistical techniques and models will be created. The area will get more and more sophisticated, and will also help other fields of astronomy.

7.3 India's contribution towards PTA development

The Indian Pulsar Timing Array (InPTA) is a highly specialised Indo-Japanese physics collaboration designed to detect **nanohertz gravitational waves**. The Indian Pulsar Timing Array (InPTA) employs unique features of the upgraded Giant Metrewave Radio Telescope (uGMRT) to monitor dozens of the International Pulsar Timing Array (IPTA) millisecond pulsars (MSPs), simultaneously in the 300–500 MHz and the 1260–1460 MHz bands. InPTA is a member of the International Pulsar Timing (IPTA) since 2021 and is bound to release data to the public and the collaboration. As of now, the InPTA has published two data releases.

The first one in 2022, consisting of the data from the last three to four years. It included high-precision time-of-arrival (TOAs) data and interstellar dispersion measures for 14 millisecond pulsars. The second data release was done in 2025, consisting of a timeline of continuous 7 years of data. This one had the data of 27 millisecond pulsars.

The Indian PTA is turning out very positively, as the international collaboration is expected to publish a third data release.

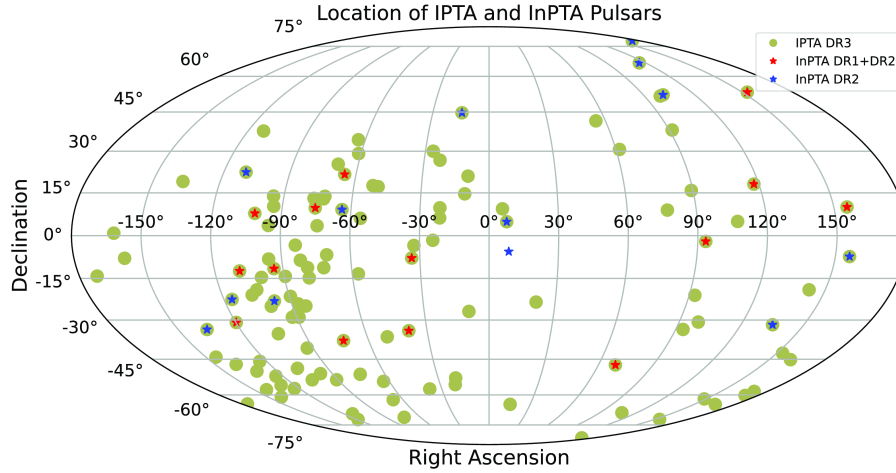


Figure 7.1: The sky distribution for 27 pulsars included in this data release is shown, marked by red and blue stars, representing observations made with the InPTA experiment between November 2016 and March 2024. 14 pulsars indicated by red stars were part of the InPTA DR1, whereas pulsars marked by blue stars are added in the present data release along with 14 InPTA DR1 pulsars. Green circles indicate pulsars that are planned to be included in the upcoming third data release of IPTA.

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